

Systematic evaluation of intratumoral and peripheral BCR repertoires in three cancers

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Abstract

The current understanding of humoral immune response in cancer patients suggests that tumors may be infiltrated with diffuse B cells of extra-tumoral origin or develop organized lymphoid structures, where somatic hypermutation and antigen-driven selection occur locally. These processes are thought to be significantly influenced by the tumor microenvironment in the form of secretory factors and biased cell-cell interactions. To address the manifestation of this influence, we used deep unbiased immunoglobulin profiling and systematically characterized the relationships between B cells in circulation, draining lymph nodes (draining LNs), and tumors in 14 patients with three human cancers. We show that draining LNs are differentially involved in the interaction with the tumor site and that there is significant heterogeneity even between different parts of a single lymph node (LN). Next, we confirmed and elaborated upon previous observations of intratumoral immunoglobulin heterogeneity. We identified B cell receptor (BCR) clonotypes that were expanded in tumors relative to draining LNs and blood and observed that these tumor-expanded clonotypes were less hypermutated than non-expanded (ubiquitous) clonotypes. Furthermore, we observed a shift in the properties of complementarity-determining region 3 of a BCR heavy chain (CDR-H3) towards less mature and less specific BCR repertoire in tumor-infiltrating B-cells compared to circulating B-cells, which may indicate less stringent control for antibody-producing B cell development in tumor microenvironment (TME). In addition, we found repertoire-level evidence that B-cells may be selected according to their CDR-H3 physicochemical properties before they activate somatic hypermutation (SHM). Altogether, our work outlines a broad picture of the difference in the tumor BCR repertoire relative to non-tumor tissues and points to the unexpected features of the SHM process.

eLife assessment

This study provides **useful** insights into inter- and intra-site B cell receptor repertoire heterogeneity, noting that B cell clones from the tumour interact more with their draining lymph node than with the blood and that there is less mutation/expansion/activation of B cell clones in tumours. Unfortunately, the main claims are **incomplete** and only partially supported. The work could be of interest to an audience including medical biologists/immunologists and computational biologists across cancer specialities.

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Introduction

B cells are a relatively undercharacterized component of the tumor microenvironment. In human tumors, infiltration with B-cells, as cells and T-cells is often a positive prognostic factor, especially in conjunction with the formation of tertiary lymphoid structures (TLSs). Mechanistically, tumor-infiltrating B-cells (TI-Bs) may act via presentation of BCR-cognate antigens to T-cells^{1–3}, production of pro- or anti-tumor cytokines^{4–7}, and production of antibodies, which may be tumor-specific⁸ and enhance killing of tumor cells via antibody-dependent cytotoxicity (ADCC)⁹ and complement-induced cytotoxicity, enhance antigen-presentation by dendritic cells¹⁰ or form immune complexes that promote the activation of myeloid-derived suppressor cells (MDSC)¹¹. The effector functions of antibodies are highly dependent on antibody isotype¹² and specificity. For instance, IgG1 is the most efficient factor in ADCC, and IgG3 (together with IgG1) is capable of complement-dependent cytotoxicity¹³. These functions depend on the ability of an antibody to bind its cognate antigen, which increases during B-cell maturation and somatic hypermutation¹⁴. Isotype switching, SHM, and B-cell proliferation and differentiation are influenced by the tumor microenvironment, neoantigen burden, and presence of certain driver mutations¹⁵. Therefore, the properties of immunoglobulins produced within the tumor are both a reflection of TME features and a factor that shapes the TME.

The whole population of immunoglobulin receptors and immunoglobulins produced within a certain tissue or cell population is known as a BCR repertoire. BCR repertoire analysis has become an important approach for characterizing adaptive immune responses in health and disease^{16–18}. Starting from genomic DNA or RNA, libraries representing the sequences of variable domains of antibodies can be produced and sequenced using high throughput sequencing^{19–23}. Alternatively, immune receptor sequences can be derived from RNA-Seq data^{24–27}. Then, the properties of the variable parts of immune receptor sequences are studied using bioinformatics tools that are currently quite elaborately developed^{28–34}.

For BCR repertoires, features that are considered functionally important are clonality, isotype composition, biases in V- and J-gene usage, and extent of SHM^{35–38}. In addition, characteristics of the immunoglobulin CDR-H3 region, such as the number of added nucleotides, length, and amino acid physicochemical properties, have functional implications and have been linked to antibody specificity³⁹ and to the B-cell maturation stage⁴⁰.

Correspondingly, the characteristics of the immune repertoire have been found to be associated with clinical outcomes^{15,25,41–43}. Specifically, high intratumoral immunoglobulin heavy chain (IGH) expression, high IGH clonality, and a high proportion of IgG1 among all IGH transcripts were strongly correlated with higher overall survival in melanoma²⁵. Similarly, high

IgG1 mRNA levels are positively associated with improved prognosis in early breast cancer⁴⁴. A high intratumoral IgG1 to IgA ratio was associated with improved overall survival in bladder cancer⁴⁵ and for *KRAS*mut but not *KRAS*wt lung adenocarcinoma (LUAD), suggesting the first link between driver mutation and B-cell response¹⁵. In lung adenocarcinoma, breast cancer, and bladder cancer, only a subset of V-segments is associated with improved survival⁴⁶. The co-occurrence of specific antibody motifs and mutated tumor-derived peptides, presumably indicating specificity to particular tumor neoantigens, was also correlated with longer survival in colorectal cancer⁴³.

The tumor immune repertoire may be used as a source of tumor-antigen-specific antibodies for therapy development^{47–49}. For instance, in melanoma 30%–80% of the total BCR repertoire²⁵ may constitute one or several dominant B cell clones. Corresponding antibodies can be produced, verified for tumor reactivity⁴⁹, and further employed in chimeric antigen receptor T cell (CAR-T) therapy or other therapeutic approaches. The knowledge of antigen specificity of cancer-associated antibodies is currently insufficient, and it is mostly limited to autoantigens and some tumor-associated antigens^{50–55}.

However, even in the absence of knowledge of cognate antigens, certain hypotheses may be derived from BCR repertoire characteristics. Some individual IGHV-genes and subgroups have been associated with autoimmunity⁵⁶. Likewise, an increase in the BCR clonal expansion index⁵⁷, dominated by IgA and IgM isotypes, was associated with systemic lupus erythematosus (SLE) and Crohn's disease³⁸.

The ability to accurately characterize the B cell repertoire in the tumor microenvironment is of vital importance for both fundamental and clinical challenges. However, tumor tissue may not always be available for analysis; therefore, it would be beneficial to derive information about the tumor BCR repertoire from peripheral blood or draining LNs. It is not known, however, whether tumor-dominant BCR clonotypes can be found in the peripheral blood of cancer patients and at what frequencies. The ability to detect these clonotypes in a patient's peripheral blood could have a predictive value for disease prognosis. In addition, it is not known how the BCR repertoire characteristics of peripheral blood B-cells and tumor-draining LNs relate to the repertoire of TI-Bs.

Therefore, in the current study, we aimed to comprehensively and systematically evaluate tumor, lymph node, and peripheral blood BCR repertoires and their interrelationships. We show that the tumor BCR repertoire is closely related to that of draining LNs, both in clonal composition and isotype proportion. Furthermore, we observed that different LNs from one draining lymph node pool may be differentially involved in the interaction with the tumor, as reflected by the similarity of the BCR repertoire clonal composition. CDR-H3 properties indicate a less mature and less specific BCR repertoire of tumor-infiltrating B cells compared to circulating B cells. BCR clonotypes that expand in a tumor relative to other tissues are, on average, less hypermutated than non-expanded (ubiquitous) clonotypes.

Results

Experimental and Computational Study Design

To systematically study the relationships between BCR repertoires in tumors and normal peripheral compartments, we performed RNA-based targeted BCR repertoire analysis from four tissue types: tumor (tum), corresponding normal tissue (norm), tumor-draining LNs, and peripheral blood mononuclear cells (PBMC), of 14 cancer patients (melanoma, *n* = 6; lung cancer, *n* = 4; and colorectal cancer, *n* = 4). To account for spatial heterogeneity, we obtained 3 fragments of tumor tissue per patient. For draining LNs, we either dissected them into three separate pieces to study intra-LN spatial heterogeneity (lung cancer and melanoma, parts of LNs designated as LN11,

LN12, LN13), or, where available, obtained three separate draining LNs (designated as LN1, LN2, LN3) (**Fig. 1A**). All fragments were homogenized separately into single-cell suspensions. To account for sampling noise at the level of individual cells, we obtained two replicate samples from each fragment after homogenization. The relative similarity of clonotype frequencies in replicate samples and separate tumor fragments is shown in **Fig. 1B**. BCR repertoires were obtained using the 5'-RACE (Rapid Amplification of cDNA Ends) protocol as previously described²¹.

In this study, we used bulk RNA-based BCR profiling, with the understanding that the dominant clonotypes and B cell lineages reproducibly represented in biological replicates reflect the presence of clonally expanded plasma cells, plasmablasts, and the most expanded memory B cell clones. In other words, the RNA-based approach mainly revealed the repertoire of functionally active B-cell lineages (Supplementary Note 1).

In most experimental setups, we employed short 150+150 nt sequencing covering the IGH CDR-H3 regions, using 5'RACE method. This approach allowed us to achieve high coverage for the obtained libraries (Table S1) to reveal information on clonal composition, CDR-H3 properties, IgM/IgG/IgA isotypes, somatic hypermutation load within CDR-H3, and clonal B cell lineages.

Tumor/non-tumor repertoire overlap and isotype composition

First, we characterized the relative similarity of IGH repertoires derived from tumors, tumor-draining LNs, and PBMC on the individual CDR-H3 clonotype level. We define clonotype as an instance with an identical CDR-H3 nucleotide sequence and identical V- and J-segment attribution (isotype attribution may be different). Unlike other authors, here we do not pool together similar CDR-H3 sequences to account for hypermutation. (Hypermutation analysis is done separately and defined as clonal group analysis.)

As overlap metrics are dependent on overall repertoire richness, we normalized the comparison using the same number of top most frequent clonotypes of each isotype from each sample ($N = 109$). Repertoire data for each sample were split according to the immunoglobulin isotype, and the F2 metric was calculated for each isotype separately and plotted as an individual point. We used the repertoire overlap metric F2 (Clonotype-wise sum of geometric mean frequencies of overlapping clonotypes), which accounts for both the number and frequency of overlapping clonotypes (**Fig. 2A**). As expected, significantly lower overlaps were observed between the IGH repertoires of peripheral blood and tumors compared to LN/tumor overlaps. The LN/PBMC overlap also tended to be lower, but the difference was not statistically significant. We also analyzed D metric, which represents the relative overlap diversity uninfluenced by clonotype frequency ($D_{ij} = d_{ij} / (d_i * d_j)$, where d_{ij} is the number of clonotypes present in both samples, while d_i and d_j are the diversities of samples i and j respectively). The results for D metric are not shown, as they indicate a similar trend to that of F2 metric. This observation allows us to conclude that tumor IGH repertoires are more similar to the repertoires of tumor-draining LNs than to those of peripheral blood, both if clonotype frequency is taken into account, and when it is not.

These results are corroborated by visualization at the individual patient level, using Cytoscape network visualization platform to visualize the structure of the repertoire overlap. As exemplified by melanoma patient mp3 (**Fig. 2B**), the repertoire from the tumor is closely related to the tumor-draining LN repertoire, whereas the PBMC repertoire has very few overlapping clonotypes with both tumors and draining LNs. Overall, these analyses revealed that the extent of clonal exchange between tumors and PBMC was significantly lower than that between tumors and draining LNs. The frequencies of overlapping clonotypes were also more strongly correlated between tumors and draining LNs than between tumors and peripheral blood (**Fig. S2A**, R metric). The level of clonal exchange between tissues was dependent on isotype (**Fig. 2E, F**). LN/tumor overlap was higher in the IgG repertoire (**Fig. 2E**), whereas PBMC/tumor overlap was lower in the IgG repertoire (**Fig. 2F**) than in the IgA repertoire. This suggests that tumor-infiltrating IgG-expressing B-cells (IgG-TI-Bs) avoid systemic circulation, whereas IgA-expressing tumor-

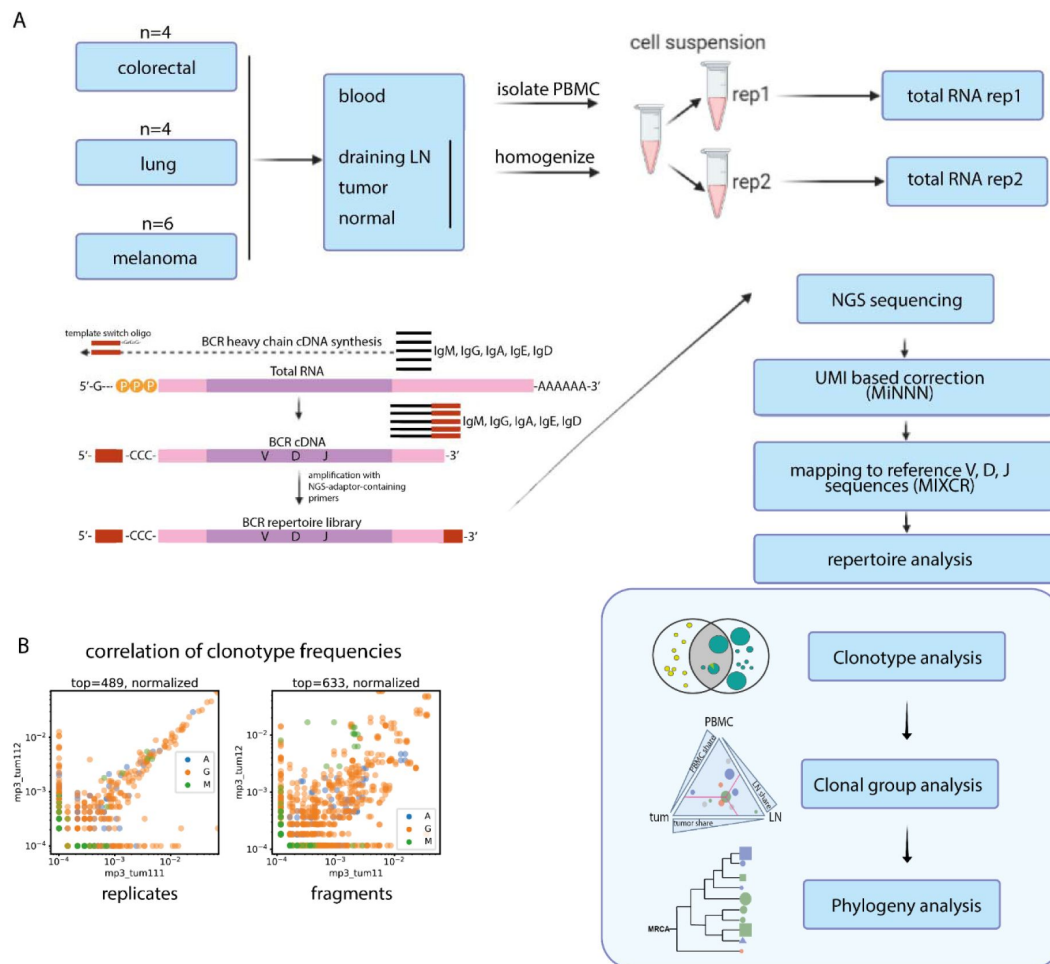


Figure 1.

Experimental design.

A - Overview of the experimental design and procedures. **B** - Using technical replicates at the cell suspension level to account for sampling bias in the clonotype frequencies.

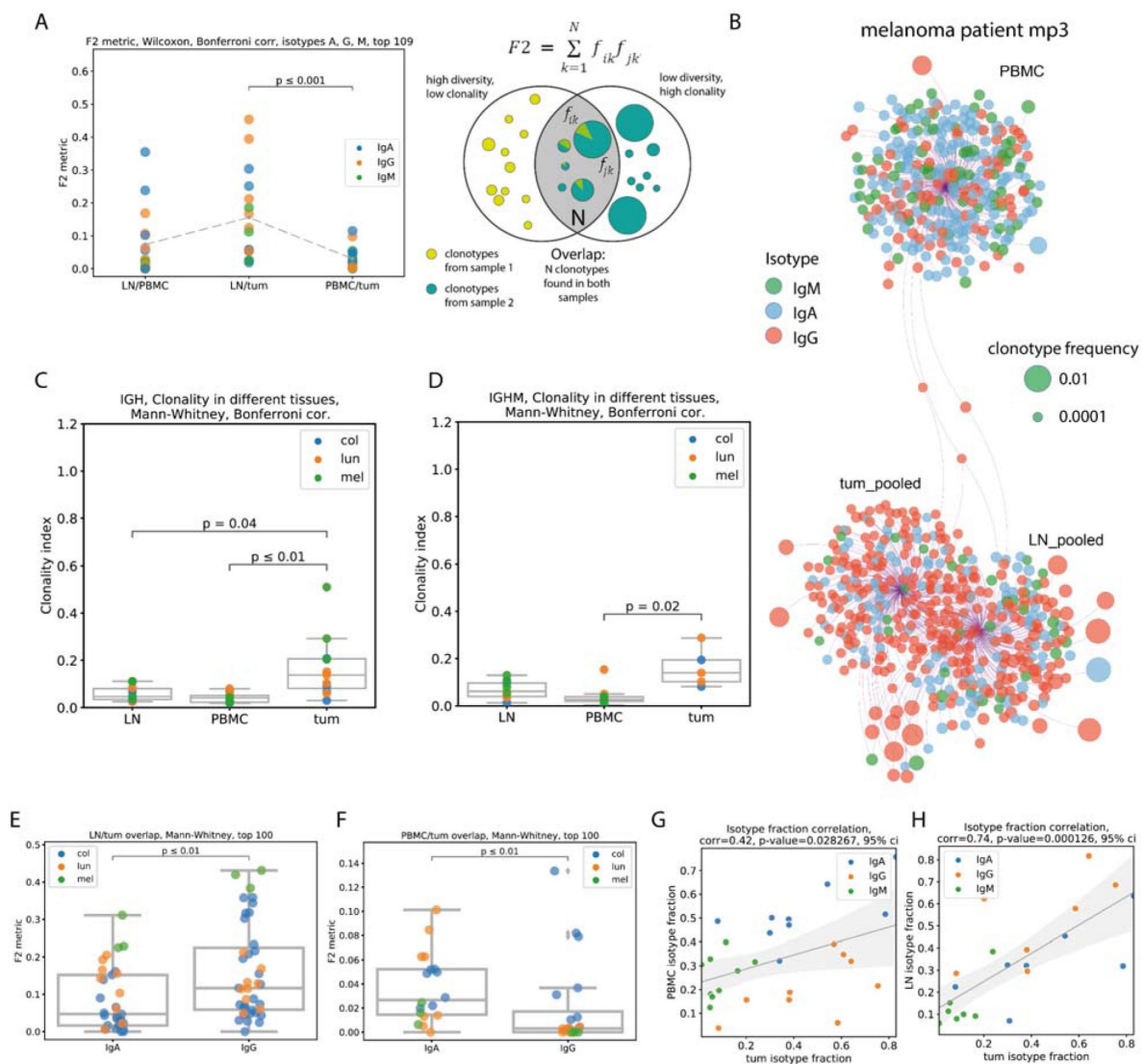


Figure 2.

Repertoire overlap, clonality and isotype composition (clonotype level, pooled patient data (except B, data from melanoma patient 3)).

A - repertoire overlap between pairs of tissues, by F2 metric, repertoires split by immunoglobulin isotype, (N=6); **B** - network representation of Ig repertoires from PBMC, tumor-draining LN, and tumor of mp3 patient (melanoma); individual clonotypes of the same origin (PBMC, tumors, or draining LN) are shown as bubbles connected with the edges to one anchor node. Clonotypes shared between tissues were connected with two or three edges to the corresponding anchor nodes and were located between them. The size of the bubbles represents the relative frequency of clonotypes within a sample, and the color represents the isotype. The relative distance between anchor nodes corresponded to the similarity of repertoires (the number of shared clonotypes). **C**, **D** - Clonality of Ig repertoires in PBMC, draining LNs, and tumors of 14 cancer patients. This reflects the presence of clonal expansion. Calculated as in [58](#); [1-normalized Shannon-Wiener index]; **C**-total IG repertoire; **D**-IgM repertoire; **E**-LN/tumor overlap for IgA and IgG repertoires (N=7); **F**- PBMC/tumor overlap for IgA and IgG repertoires (N=9); **G**, **H**-isotype fraction correlation between PBMC and tumor repertoires (**G**, N=9), or between LN and tumor repertoires (**H**, N=7).

infiltrating B-cells (IgA-TI-Bs) may be found in the peripheral blood with a higher probability. In addition, the tumor repertoire was significantly more clonal (clonality calculated as [1-normalized Shannon-Wiener index]⁵⁸) than the PBMC or draining LN repertoire, both overall (**Fig. 2C**) and separately for the IGHM repertoire (**Fig. 2D**).

We did not find a statistically significant difference in isotype composition between cancers in any of the studied tissues, with the exception of IgM percentage in melanoma tumors. In BCR repertoires from melanoma tumors, the total percentage of repertoire consisting of IgM clonotypes was significantly lower than that in other types of cancers (colorectal and lung) (Fig. S2B). The isotype composition of non-tumor tissues correlated with the isotype composition of the tumor; this effect was less prominent in peripheral blood ($R = 0.42$, $p = 0.028$) (**Fig. 2G**) and more prominent for LN ($R = 0.74$, $p < 0.01$) (**Fig. 2H**).

CDR-H3 properties

Analysis of averaged CDR-H3 repertoire characteristics revealed increased CDR-H3 length in tumors compared to PBMC for the total repertoire (**Fig. 3A**) and also in IgA and IgG repertoires separately (Fig. S3A, B), but this was not the case for IgM (Fig. S3C). In addition, the increase in CDR-H3 length in IgA repertoires from tumor-draining LNs compared to PBMC was statistically significant (Fig. S3A).

Interestingly, the only significant difference we found when comparing CDR-H3 lengths between cancer types was reduced IgA CDR-H3 length in colorectal cancer, especially compared to melanoma ($p=0.02$) (**Fig. 3B**). This could reflect a generally more mature IgA repertoire in colon tissues owing to the previous history of interactions with microbiota^{59,60}. However, the relationship between this difference and tumor antigen specificity remains to be verified.

To explore CDR-H3 physicochemical properties, we calculated the mean charge, hydrophathy, predicted interaction strength, and Kidera factors 1 - 9 (kf1-kf9) for five central amino acids of the CDR-H3 region for the 100 most frequent clonotypes of each sample using VDJtools. Kidera factors are a set of scores which quantify physicochemical properties of protein sequences⁶¹. 188 physical properties of the 20 amino acids are encoded using dimension reduction techniques, to yield 9 factors which are used to quantitatively characterize physicochemical properties of amino acid sequences. Comparing between tissues, we found that kf4 value for the tumor repertoire was decreased compared to PBMC in the total repertoire, and kf5 was decreased in tumor vs. PBMC in the IgG repertoire (**Fig. 3C**). In addition, kf6 value was decreased in the LN repertoire compared to PBMC. Kf4 inversely correlates with hydrophobicity, indicating a higher proportion of hydrophobic residues in BCR CDR-H3s from the tumor repertoire. Kf5 reflects a double-bend preference and has not been previously found to be significant in the context of antibody properties. Kf6 is a measure of partial specific volume; therefore, a lower kf6 value indicates less bulky amino acid residues in CDR-H3s from the LN repertoire. Between the tumor and normal tissue repertoires, the hydrophathy value was lower in normal lung tissue, also indicating a higher proportion of hydrophobic residues in tumor-derived CDR-H3 repertoires (**Fig. 3D**). According to⁴⁰, more mature B-cell subpopulations have higher mjenergy, disorder, kf4, kf6, and kf7 and lower CDR-H3 length, strength, volume, kf2, and kf3. Again, the mean CDR-H3 charge was negatively associated with specificity⁶² and beta-sheet propensity was associated with antibody promiscuity⁶³. Poly-reactive and self-reactive antibodies have, on average, longer CDR-H3s⁶⁴ with a higher charge⁶² and net hydrophobicity⁶⁵⁻⁶⁷.

Therefore, collectively, our observations suggest a less mature and less specific BCR repertoire of tumor-infiltrating B cells compared to circulating B cells and B cells infiltrating normal tissue, which may indicate less stringent control for antibody-producing B cell development in the TME.

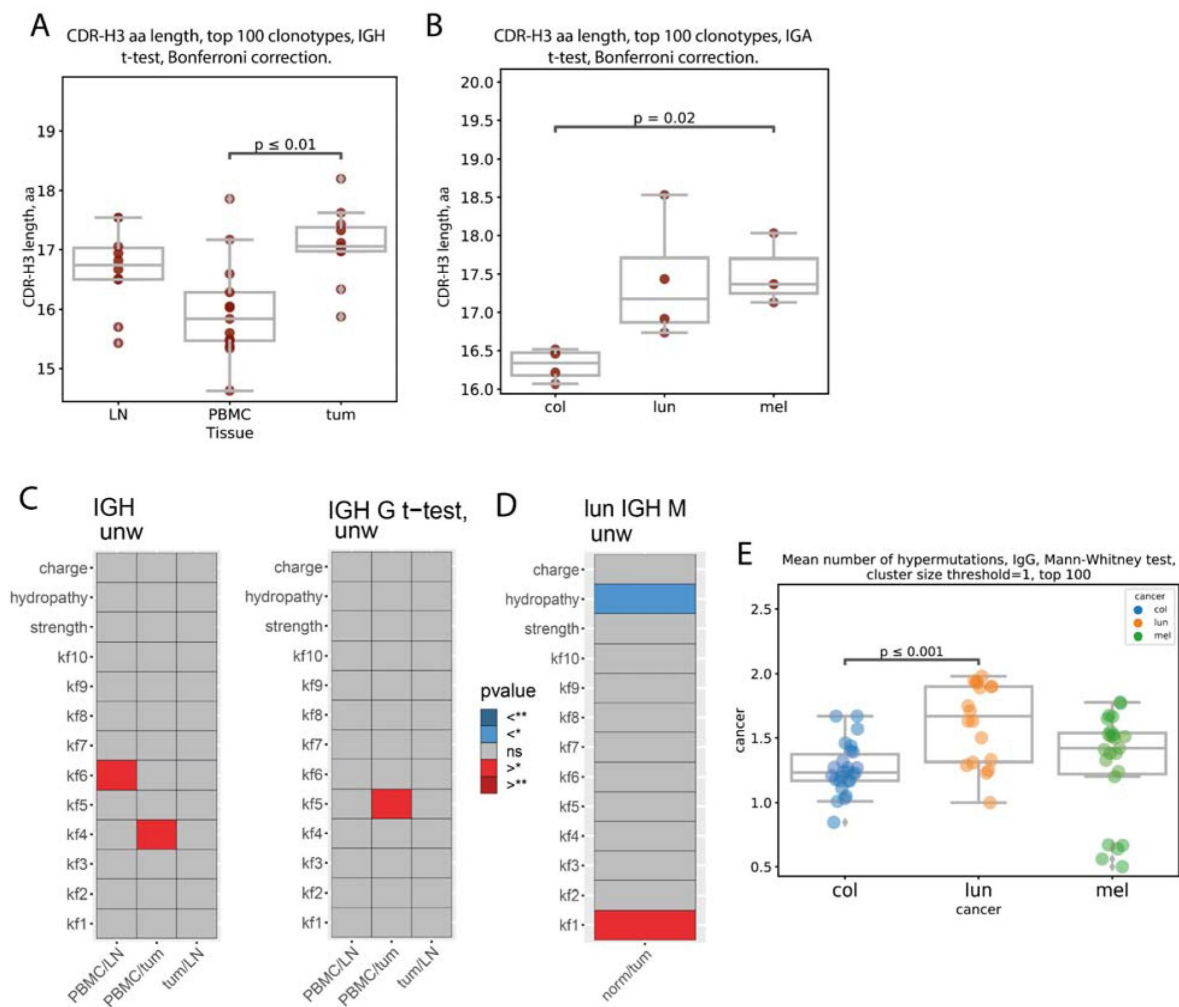


Figure 3.

CDR-H3 amino acid properties (clonotype level, pooled patient data)

A - Mean amino CDR-H3 length of top 100 most frequent clonotypes from tumor, lymph node and PBMC tissues irrespective of isotype CDR-H3 are on average significantly longer in tumor than PBMC for total repertoire (N=14, $p < 0.01$, two-sided t-test, Bonferroni correction); **B** - Comparison of mean amino acid CDR-H3 length of 100 most frequent clonotypes for colon, lung and melanoma cancer samples from, tumor. CDR-H3s of tumor-infiltrating clonotypes were shorter for colorectal cancer patients compared to melanoma in IgA repertoires (N=11, $p = 0.02$); **C** - Comparison of amino acid properties in the central region of CDR-H3, for total repertoire (**C-left**) or IGHG repertoire (**C-right**), all cancers (significantly increased - **red**, significantly decreased - **blue**) two-sided t-test, Bonferroni-Holm correction; **D** - Comparison of amino acid properties in the central region of CDR-H3, for IGHM repertoire, lung cancer (significantly increased - **red**, significantly decreased - **blue**); ($p < 0.01$, two-sided t-test, Bonferroni-Holm correction); **E** - Average number of mutations relative to germline for tumor samples from different types of cancers, N=14.

Immunoglobulin hypermutation analysis across tissues and isotypes

The intensity of somatic hypermutation (average number of mutations relative to the most recent common ancestor, MRCA) reflects the average extent of antigenic selection experience of the clonotypes found in a given tissue or cancer type. No significant difference was found between PBMC, tumor-draining LNs, and tumors for the top 100 most frequent clonotypes in the total repertoire as well as for the top 100 most frequent clonotypes of each isotype separately (**not shown**). This indicates that in the RNA-based BCR repertoires, in all studied tissues, the most dominant immunoglobulin clonotypes belong to cell populations with an equivalent history of antigen exposure, selection, and maturation. However, there was a statistically significant difference in the number of hypermutations in IgG between the cancers (**Fig. 3E** [↗](#)). IGHG clonotypes from lung cancer samples show higher number of hypermutations, possibly reflecting high mutational load found in lung cancer tissue. For melanoma, another cancer known for high mutational load, no statistically significant difference was found. This may be due to higher variance between melanoma samples, which hinders the analysis, or due to the small sample size.

Clonal exchange between tissues at the level of B cell lineages

Next, we investigated clonal exchange between the PBMC, tumor-draining LNs, and tumors at the level of hypermutating clonal lineages, which are likely to be involved in recent and ongoing immune responses. The results are shown in **Fig. 4A** [↗](#). After pooling patient-wise clonotype data, clonal groups were formed by sequence similarity, and then IGH clonotypes within a group were assembled into clonal lineages that shared a common ancestor⁶⁸ [↗](#) and represent a B-cell clone undergoing the affinity maturation process. Each clonotype within a clonal group was attributed to the tissue of origin, tumor, LN, and/or PBMC, and to a particular isotype. For each clonal group, the percentage of clonotypes belonging to each isotype and tissue was calculated. Clonal groups from all patients with a given cancer type were plotted on a triangle plot using the percentage of clonotypes from the tumor, LN, and PBMC as coordinates and colored according to the dominant isotype (>60%).

In all studied cancer types, IgA-dominated clonal groups were evenly distributed among the three tissues, indicating no preference for IgA-switched cells towards lymphoid or tumor tissue residence. IgG-dominated clonal groups showed a preference for tumor and LN residence in lung cancer and melanoma, in accordance with the idea of tight interaction between LN and tumor in mounting anti-tumor immune responses. IgM-dominated clonal groups showed a strong preference for colorectal cancer tumors, indicating intensive intratumoral somatic hypermutation without isotype switching (**Fig. 4B**, **red triangle** [↗](#)).

LN-to-LN heterogeneity in colorectal cancer

Next, we sought to investigate whether BCR repertoire analysis could discern among a group of tumor-draining LNs that are in more intensive clonal exchange with the tumor. This question may be addressed at the individual clonotype level and at the level of hypermutating clonal groups.

At the individual clonotype level, we compared F2 metric values for pairwise tumor/LN BCR repertoire overlaps from 2 colorectal cancer patients, for which we obtained three separate LNs from the excised surgical material. For patient ccp2, a significantly higher overlap of tumor repertoire with one of the LNs was observed (**Fig. 5 A** [↗](#)). Similarly, Cytoscape network analysis showed more clonotype sharing between LN3 than between other LNs (**Fig. 5 B** [↗](#)). For ccp3, no significant difference was observed between LNs (**Fig. 5 A**, **right panel** [↗](#)). Similarly, an unequal interaction of tumors with LNs was observed at the level of hypermutating clonal groups. We used the proportions of clonotypes originating from LN1, 2 and 3 among all LN-clonotypes within a clonal group as coordinates for triangle plots (**Fig. 5 C** [↗](#)). For ccp2, most of the LN3-focussed

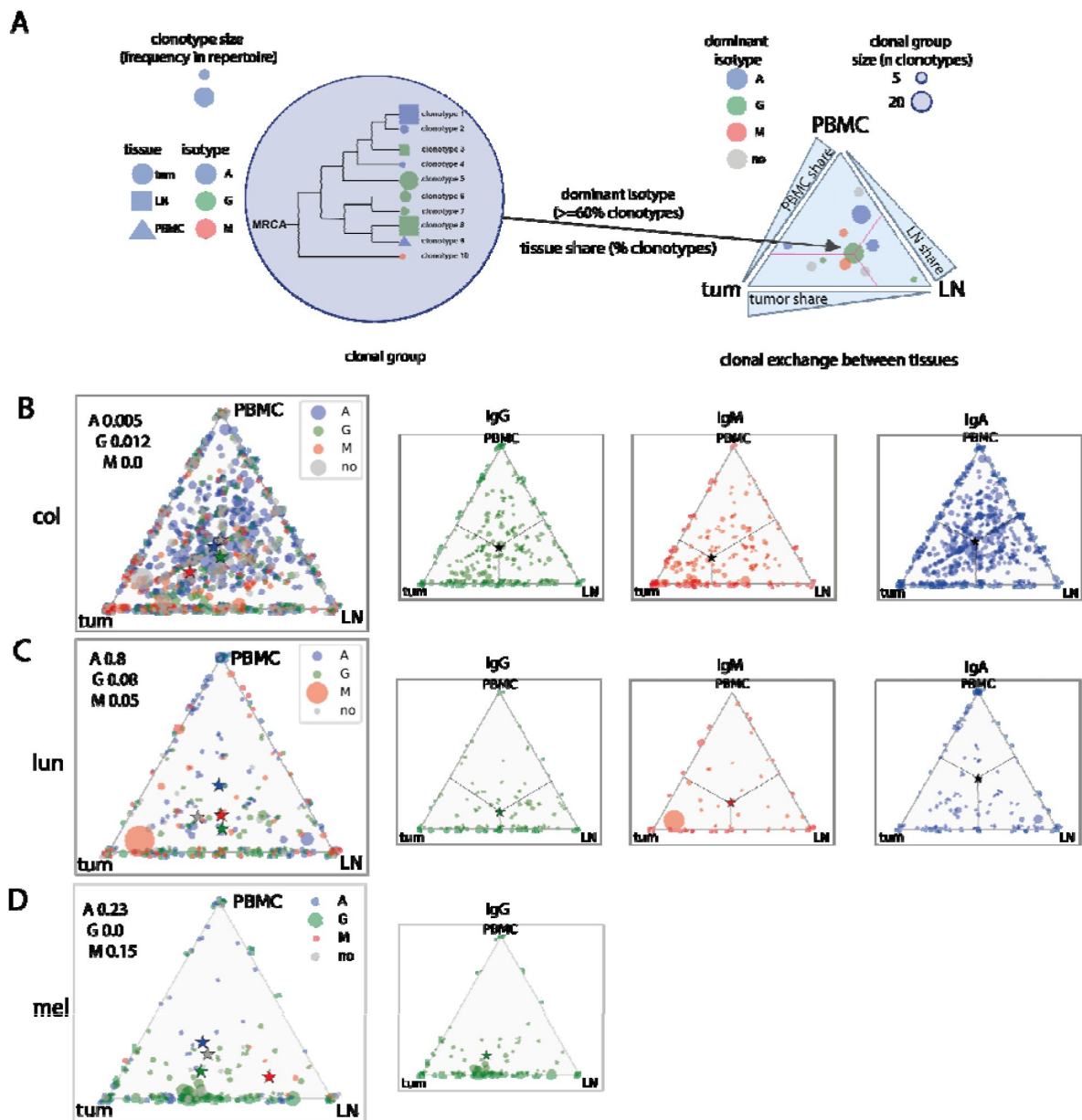


Figure 4.

Immunoglobulin hypermutation analysis across tissues and isotypes (clonal group level pooled patient data)

A - Schematic representation of analysis and triangle plot visualization of clonal group distribution between tissues; **B, C, D** - clonal group distribution between tissues for colorectal (**B**), lung (**C**), and melanoma (**D**); stars represent **non-weighted by size mean** center of triangle coordinates. Chi-2 test for goodness of fit was used to test whether each tissue equally contributed to clonal group formation.

clonal groups were also prominently represented in the tumor. However, most of the other clonal groups resided in the center of the triangle, so the mass center was not shifted towards any of the LNs overall. Functionally, this may again indicate that within a group of LNs, nodes are unequal in terms of access to tumor antigens, and this inequality shapes the BCR repertoires within these LNs. On **Fig. 5D** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)

These observations reflect a complex interplay between tumors and LNs, which may be involved in the adaptive immune response to tumor antigens.

Intra-LN heterogeneity

Likewise, we asked whether parts of LNs are equally involved in interaction with tumor. For lcp3, we obtained BCR repertoires from 3 fragments of one draining LN and compared them to tumor repertoires on the individual clonotype and clonal group level. Repertoire from fragment LN11 showed significantly lower overlap with the tumor than other two LN fragments, LN12 and LN13 (**Fig. 6A, left** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)). Triangle plot analysis of clonal group distribution also showed that the majority of the tumor-dominated (red) clonal groups resided in the LN12-LN13 side of the triangle (**Fig. 6B, left** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)).

Intratumoral heterogeneity of immune repertoires

Spatial heterogeneity of the T cell receptor repertoire is known to reflect the mutational landscape of tumor ⁴¹[we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)⁶⁹[we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#). Statistically, the magnitude of intratumoral genetic heterogeneity correlates with the heterogeneity of immune cell infiltration, implying the co-evolution of the tumor genetic architecture and immune microenvironment ⁷⁰[we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#). Higher heterogeneity of tumor-infiltrating T-cell repertoire is associated with higher risk of recurrence and shorter disease-free survival ⁴¹[we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#).

B-cell repertoire heterogeneity, however, remains relatively understudied. The mechanisms underlying TI-Bs heterogeneity may involve not only differential infiltration and accumulation of lymphocytes, but also local development of tertiary lymphoid structures which then drive the development of local immune response specific to subclonal neoantigens.

Here we aim to quantify intratumoral BCR repertoire heterogeneity, measured as the extent of clonotype or clonal group sharing between parts of the tumor. First, we confirm for all studied tumor samples, that repertoires derived from separate tumor fragments are significantly less overlapping (F2 metric) than repertoires from biological replicates of one tumor fragment (Fig.S4). Then we pooled repertoires from replicates belonging to the same tumor fragments and analyzed the overlap between top 300 clonotypes from fragments of the same tumor using Cytoscape platform (**Fig. 6C** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)). We observed significant patient-to-patient variability in the degree of heterogeneity between tumor fragments. In a tumor from patient **ccp2**, all three fragments have a high number of clonotypes in common (represented also by close distance between anchor nodes on the bubble diagram), and the isotype composition is also similar and dominated by IgA (**Fig. 6C, D left** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)). Overall, this draws a picture of a relatively homogenous immune infiltrate in the tumor of this patient. In patient **ccp3** (**Fig. 6 C, D, right** [we show an example of a clonal group that includes clonotypes from tumor \(multiple tumor fragments\) and only from one of the LNs, to illustrate asymmetric involvement of LNs in the development of a hypermutated clonotype. For ccp3, no such inequality was observed. To validate this hypothesis, it would be beneficial to obtain a BCR repertoire from non-tumor-draining LNs; however, this was not possible in the current study.](#)), we see a remarkably different pattern. In ccp3, fragment **tum_1** is dominated by IgA and has many shared clonotypes with fragment **tum_3**, which are also IgA-clonotypes.

Fragment **tum_2** shares many clonotypes with **tum_3**, and these clonotypes, as also total repertoires of fragments **tum_2** and **tum_3**, are mostly IgG and IgM clonotypes. Heterogeneity of clonal group distribution was also significantly more prominent in **ccp3** compared to **ccp2**

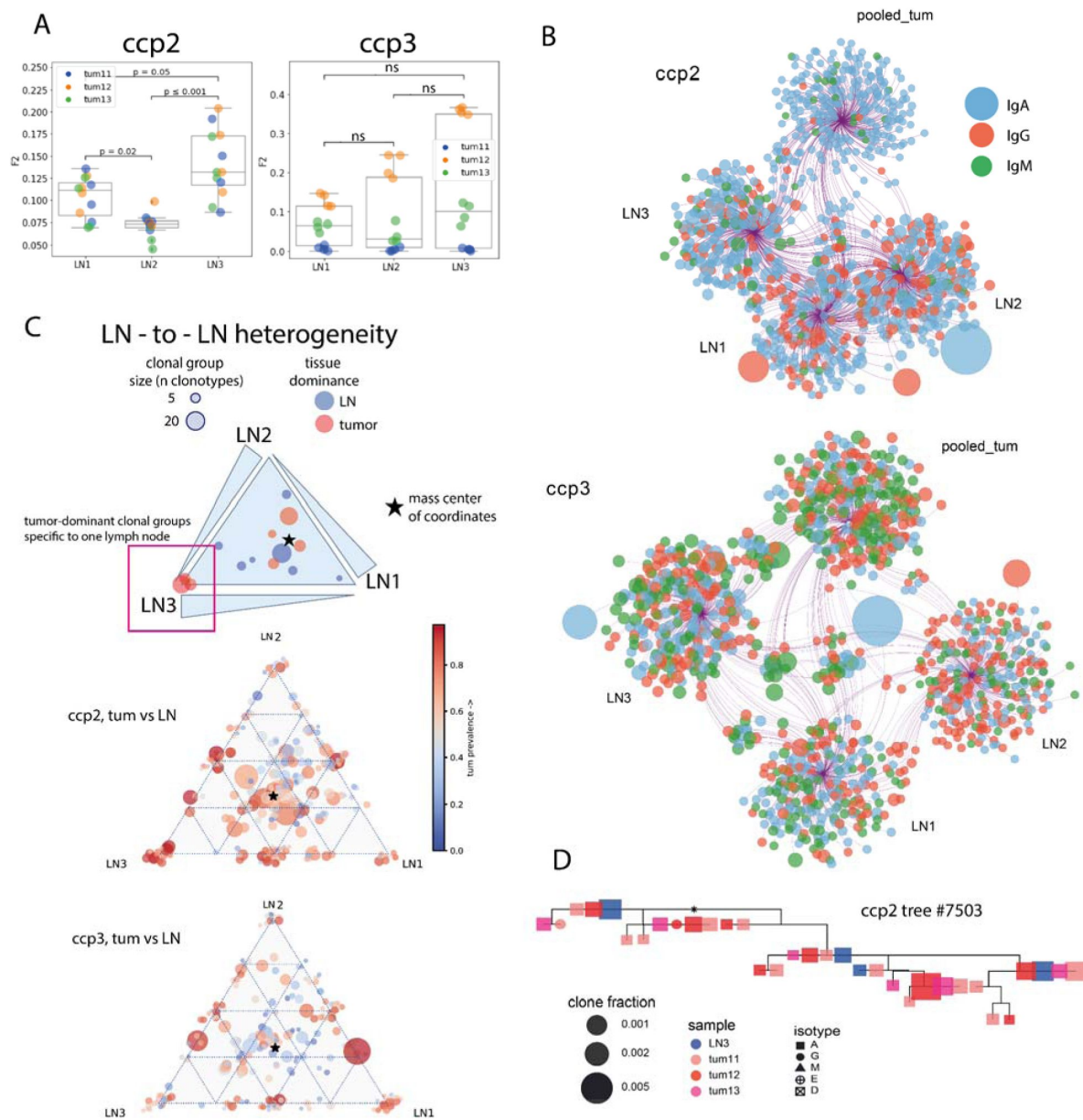


Figure 5.

LN-to-LN difference of BCR repertoires in colorectal cancer (clonotype level - panels A, B; clonal group level - panels C, D; individual patient data).

A - repertoire overlap between tumor and three separate LNs from the draining lymph node pool; Mann-Whitney test, Bonferroni correction. **B**: Network representation of Ig repertoires from tumor and three separate LNs, with circles representing individual CDR-H3 sequences, size of the circles corresponding to the clone frequency, and color corresponding to the isotype; **C** - triangle plot visualization of clonal group distribution between three different LNs, with size corresponding to the number of individual CDR-H3 sequences (clonotypes) within a given clonal group, and color corresponding to the percentage of tumor-derived clonotypes-within the clonal group; **D**-example of a clonal lineage consisting of CDR-H3 sequences derived from a lymph node (blue) and all three tumor fragments (shades of red) from patient ccp2, shapes representing isotypes and size representing frequency of a given sequence in a given sample.

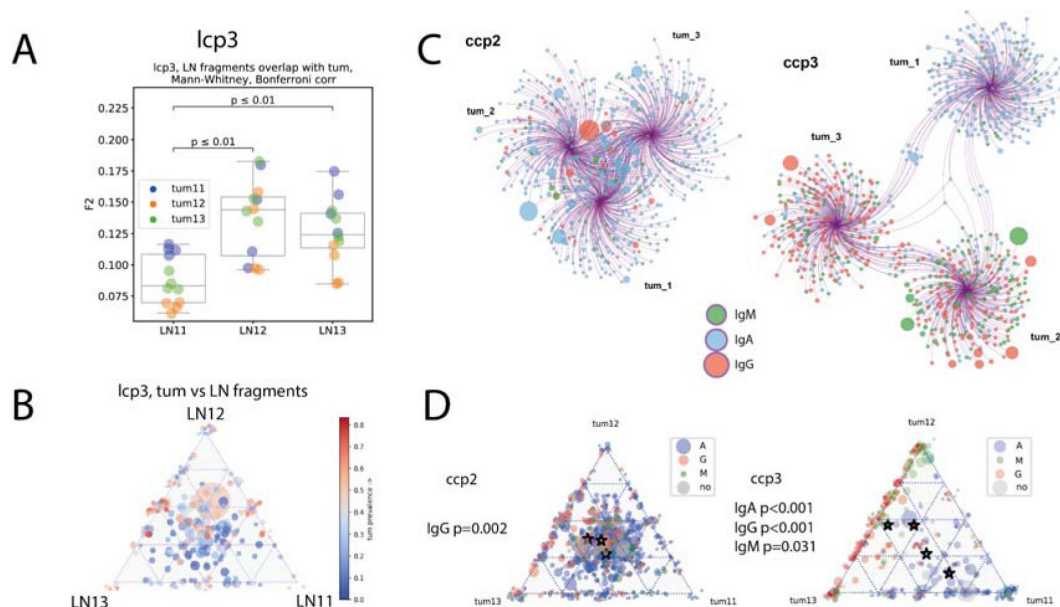


Figure 6

Intra-LN and intratumoral heterogeneity (clonotype level - panels A, C; clonal group level - panel B, D; individual patient data).

A - repertoire overlap between tumor and lymph node fragments for colorectal cancer patient ccp6 and lung cancer patient lcp3; Mann-Whitney test, Bonferroni corr. B - triangle plot visualization of clonal group distribution between lymph node fragments, with size corresponding to the number of individual CDR-H3 sequences (clonotypes) within a given clonal group, and color corresponding to the percentage of tumor-derived clonotypes within the clonal group; C - network representation of Ig repertoires from tumor fragments, with circles representing individual CDR-H3 sequences, size of the circles corresponding to the clone frequency, and color corresponding to the isotype, edges connect clonotypes to their fragment of origin; D - triangle plot visualization of clonal group distribution between tumor fragments, with size corresponding to the number of individual CDR-H3 sequences (clonotypes) within a given clonal group, and color representing dominant clonotype; Stars represent **non-weighted by size mean** center of triangle coordinates. Chi-2 test for goodness of fit is used to test if each tumor segment equally contributes to clonal groups formation.

(**Fig.6D** [↗](#)). In **ccp2**, significant eccentricity was observed only for IgG-dominated clonal groups (red star on **Fig.6D, left** [↗](#)), which tended to share between fragments tum_2 and tum_3, and not tum_1. This may indicate that IgA clonal groups are specific either to clonal tumor antigens (that is, present in all parts of tumor), or to other ubiquitous antigenic determinants, whereas IgG clonal groups more likely are specific to tumor specific and/or subclonal antigens (present only in some parts of tumor). In **ccp3**, most clonal groups show a significant degree of eccentricity on triangle plot (**Fig.6D, right** [↗](#)), indicating heterogeneous distribution between tumor fragments, for IgA-dominated as well as for IgG-dominated clonal groups. Given that clonality of ccp3 tumor fragments tum_2 and tum_3 is also higher than that of **ccp2** tumor, we hypothesize that heterogeneity plus high clonality indicate formation of TLSs and thus efficient local immune response towards tumor-related antigens.

To analyze the presence of TLSs in our tumor samples, we used histology and immunohistochemistry. We found frequent dense leukocytic accumulations in the peritumoral region, as well as distant from the tumor. The latter were the mucosal/submucosal layers, whereas peritumoral ones were found in all the intestinal layers (Fig. S5B). Such an appearance in the outer layers and prominent accumulation near the tumor indicates that these are more likely to be TLSs, rather than Peyer's patch-like structures that are localized predominantly in the mucosal and submucosal layers. In order to evaluate whether this accumulation has organized TLSs-related lymphocyte distribution, we used multicolor immunohistochemical staining of T and B cells together with TLSs-related markers: high endothelial venule marker PNAd and CXCL13 chemokine, which drives TLSs formation. We found that CD20 B cells and CD3 T cells were the major cell types within these patches, indicating their lymphoid origin (Fig S5E,F). Peritumoral structures have condensed B cells follicles surrounded by T cells and high endothelial vessels (HEV), confirming that these are primary follicular TLSs^{71,72} [↗](#). Distant lymphoid structures in the mucosal layer have a more dispersed and intermixed T and B-cell distribution, indicating a lower degree of maturation. Similar lymphoid aggregates in the mucosa and peritumoral regions have been observed on all tissue blockers for patients with colorectal cancer (two for each patient). However, their comparison is unrepresentative, owing to variations in the sampling of tissue fragments.

The design of our study included obtaining cellular-level replicates for each processed tissue fragment. This allowed us to reliably detect CDR-H3 clonotypes that were significantly expanded in a given sample relative to other samples. **Fig.7** [↗](#) shows an example of such an analysis. We used the EdgeR library from the Bioconductor package to detect clonotypes that were differentially expanded in separate fragments of tumors. These clonotypes are represented in **Fig.7 A, B** [↗](#) as colored circles on a frequency correlation plot. Expanded clonotypes with the highest frequencies were also visible on Cytoscape plots (**Fig.7 C** [↗](#)) and labeled according to their sequences. Interestingly, most expanded clonotypes are not attributed to any clonal group (and thus are not actively hypermutating), and the largest of these are of IgM isotype. Only one of the expanded clonotypes in this tumor was involved in the hypermutation process, and the structure of its clonal lineage is shown in **Fig.7 D** [↗](#). On the other hand, none of the largest clonal lineages detected in this patient contained any expanded clonotypes (**Fig.7 E** [↗](#)). In addition, on average, expanded clonotypes have fewer mutations than non-expanded clonotypes, both in tumors and in tumor-draining LNs (**Fig.7 F, all patients** [↗](#)).

Short vs long trees - phylogeny analysis

CDR-H3 is the most variable component of the BCR and is also the most important in terms of antigen recognition. However, missing mutations in other segments may lead to difficulties or a complete inability to accurately recover the phylogeny of hypermutations. Moreover, shorter sequencing reads, which may be sufficient to obtain CDR-H3 data, cover the V segment to a lesser extent, which may lead to less accurate germline gene assignment and, consequently, erroneous rooting of the phylogenetic tree.

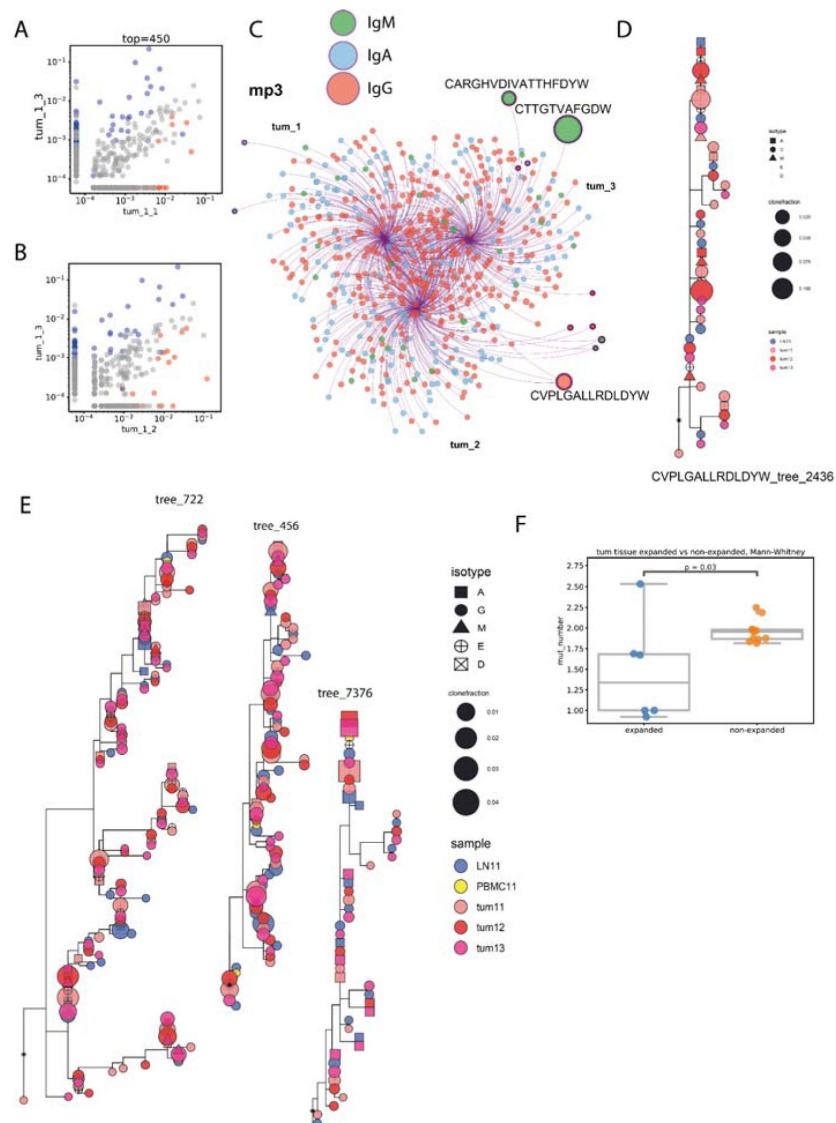


Figure 7

Expanded clonotypes and hypermutation analysis (clonotype level - A, B, C, F; clonotype group level - D, E; individual patient data).

A, B - Visualization of expanded clonotypes on frequency correlation plots for pairs of tumor fragments of melanoma tumor from patient mp3; **C** - Cytoscape network visualization of top 300 most frequent individual Ig CDR-H3 clonotypes, colored by isotype, with size of circles proportional to the frequency of a given clonotype; **D** - example of a clonal lineage containing expanded CDR-H3 sequence; **E** - examples of the biggest clonal lineages, none of which contain expanded CDR-H3 sequences; **F** - average number of mutations in expanded and non-expanded clonotypes; Mann-Whitney test; N=11.

To estimate the extent to which phylogenetic trees built on short-read data and CDR-H3 sequences alone reflected the ‘true’ phylogeny recovered from full-length (CDR1 to FR4 regions) sequences, we used two types of BCR libraries obtained from the same lung cancer patient material. Short CDR-H3 sequences were obtained from libraries prepared according to the 5’RACE protocol and sequenced with a 150+150bp read length. Long sequences were obtained from libraries prepared according to the IgMultiplex protocol and sequenced at 250+250bp read length. Then, for each dataset, we independently inferred clonal lineages and built rooted phylogenetic trees for each clonal lineage of size five or more (see Materials and Methods for more details). To compare the structures of these tree sets, we selected clonotypes with the CDR-H3 sequence and the corresponding V and J genes present in both the long and short trees. We then calculated their average ranks by distance to the root within their clonal lineages and compared the ranks (Fig.S6 A). The correlation was around 0.7 ($p < 6 \times 10^{-16}$). Therefore, phylogeny inferred based on short CDR-H3 sequences generally reflects the ‘true’ phylogeny inferred from the CDR1-FR4 repertoires. However, the average distance to the root for IgM and IgA/IgG isotypes, which have significant differences for long repertoires, did not show significance for short (CDR-H3-based) repertoires (Fig.S6 B).

One potential explanation for this could be that a larger sample size and a higher sequencing depth are required for short sequences to yield statistically significant differences. Indeed, it is impossible to avoid sporadic misplacements of short sequences in the tree when the information on hypermutations is limited. On the other hand, we clearly observed a correlation between long and short tree phylogenies. Therefore, we suppose that the reason for this contradiction is an insufficient sample size. However, there is evidence that CDR-H3 data do not always perfectly represent the ‘picture’ and should be treated with particular attention.

Grimsholm et al. in 2020 that memory B cells have, on average, higher Kidera factor 4 values and lower predicted interaction strength than naïve B-cells⁴⁰. We assumed that these features may evolve during the affinity maturation process. Therefore, we checked how the mean kf4, strength, and charge of the five central CDR-H3 amino acids depend on the clonotype’s distance to the MRCA rank on the phylogenetic tree. To ensure that there was no bias in phylogeny as a result of short sequencing length and small sample size, we used full-length sequence trees inferred from lung cancer patient lcp3 and full-length repertoire of very high sequencing depth from a healthy donor. However, we did not find any correlation between strength, charge, and **kf4** and the clonotype’s position on a tree (Fig.S6 C). One exception was a near-zero increase in charge down the tree found in the healthy donor repertoire ($r=0.04$, $p=4.9 \times 10^{-5}$), which, despite being statistically significant, is probably not mechanistically meaningful. In addition, the largest (>20 members) trees had dN/dS values < 1, indicating negative selection pressure (Fig.S6 D). Taken together, these data suggest that, in the absence of a defined time-controlled antigenic stimulus, such as vaccination, in steady-state equilibrium, only a few immunoglobulin clonal groups show definitive signs of positive selection.

Productive involvement in hypermutating lineages depends on CDR-H3 characteristics

Taking into consideration the previously found differences between memory and naïve cell features by Grimsholm et al.⁴⁰, we hypothesized that amino acid features of CDR-H3 may be selected even before the start of hypermutation and affinity maturation. To test this hypothesis, we checked how the mean kf4, strength, and charge of five central CDR-H3 amino acids depend on the clonal status of the clonotype (Fig. S7) and found the results to be tissue-specific. In particular, clonotypes not assigned to any clonal lineage (singles) had lower kf4 values, and hence higher hydrophobicity, than members of some clonal groups in the tumor and PBMC repertoires. The predicted interaction strength was higher for singles in PBMC and LNs, but not in the tumor, potentially indicating that tumor-resident clonotypes have undergone selection for CDR-H3

properties, irrespective of their involvement in SHM. Finally, clonal lineages had a higher charge in tumor repertoires, but not in the LNs or PBMC, indicating a tendency towards polyreactivity in clonotypes evolving locally under the influence of TME.

These results generally match the findings of Grimsholm et al. and support the hypothesis that selection of CDR-H3 amino acid features occurs before cells undergo SHM. Tissue differences could potentially be explained by differences in the proportions of B cell subsets in different tissues.

Discussion

We performed a multi-location study of BCR repertoires in cancer patients using an unbiased BCR library preparation technique complemented by deep sequencing. We analyzed these repertoires from two complementary perspectives. At the level of individual BCR clonotypes, we studied various repertoire features and their differences between tissues of origin, as well as repertoire heterogeneity within tumor or lymph node tissue. At the level of B-cell clonal lineages, we tracked lineage sharing between tissues, and differences in clonotype features between clonotypes that belong to some clonal lineage, and those that do not.

First, we compared BCR repertoires from PBMC and bulk tumor infiltrates and the corresponding sorted CD19⁺CD20⁺CD38^{high} plasma cells, and confirmed that the bulk RNA-based BCR repertoire is dominated by plasma cells across different peripheral compartments. This is expected, given that the IG expression level is 50-500 times higher in plasma cells than in memory B-cells²¹. However, PC has a higher IgM isotype proportion in the tumor and a lower proportion of IgG in the lymph node. The higher proportion of IgM in PCs relative to the bulk repertoire may indicate that some IgM-producing plasma cells, while having an appropriate plasmablast/PC surface marker phenotype, produce relatively low levels of antibodies, which leads to them being masked in the bulk repertoire by non-IgM non-PC B cells. A higher level of antibody production correlates with the differentiation of plasmablasts/short-lived plasma cells into long-lived plasma cells. Therefore, one could hypothesize that tumor-infiltrating IgM-expressing antibody-producing cells are skewed towards a less differentiated B-cell compartment compared to other isotypes. Similar logic applies to the IgG-producing plasma cells in the LNs: a lower proportion of IgG-clonotypes in PCs compared to the bulk repertoire indicates that some of the most frequent IgG clonotypes in the bulk repertoire belong to cells with a non-PC/plasmablast phenotype, presumably stimulated memory B-cells.

BCR repertoire overlaps and the correlation of isotype proportions of tumor and lymph node samples were higher than those of tumor and peripheral blood. A higher repertoire overlap is indicative of a higher proportion of tumor-related BCR clonotypes in the draining LNs than in the peripheral blood, which illustrates the crucial role of draining LNs in the development of the anti-tumor immune response. In addition, in accordance with previous studies, our data confirmed that the tumor BCR repertoire is more clonal than that in the lymph node or PBMC. We also showed that LNs within the same draining LN pool may be differentially involved in tumor interaction. Overall, LNs are a better source of material to study anti-tumor B-cell response than PBMC, and a better source of material for a potential cancer B-cell or antibody therapy, if tumor tissue is unavailable.

The mean CDR-H3 length of the most frequent BCR clonotypes in tumors was higher than that in other tissues. This finding may indicate that tumors tend to be infiltrated with less mature B-cells than other compartments⁴⁰, and may also indicate a higher proportion of autoreactive and polyreactive^{64,73} antibodies, which is consistent with previous research. However, in colorectal cancer, the mean CDR-H3 length in the IgA repertoire was lower than that in melanoma. This may indicate that the IgA repertoire in colorectal cancer is less subject to TME-induced biases than in other cancers studied in this study. Differences in other CDR-H3 properties, such as mean

kf3 (b-sheet preference), kf4 (inversely correlated with hydrophobicity), predicted interaction strength, and charge between different tissues, including increased central-CDR-H3 charge in the tumor compared to normal tissue, may suggest a less mature and less specific BCR repertoire of tumor-infiltrating B-cells.

Clonal groups representing hypermutating B-cell lineages are distinct in their isotypes and tissue composition. In colorectal cancer, clonal groups dominated by IgA (> 60% of IgA clonotypes) tend to be shared between tumors and blood, whereas IgG-dominated (> 60% of IgG clonotypes) clonal groups tend to be shared between tumors and draining LNs. The probability of detection of a given clone in a given location is dependent on the mRNA quantity that is produced by a particular B-cell or group of B cells (BCR expression level), as well as on the lifetime of these cells in this location (lifetime being a superposition of cells entering and leaving the tissue, and the literal lifespan of the cells and their progeny). Assuming that BCR is expressed at the same level in all members of a certain clonal population, whether it is found in the blood, LN, or tumor, the probability of detecting a clone is then only a function of the “combined lifetime” of a clone within a given tissue. Therefore, the proportion of clones within a clonal group derived from a certain tissue is a snapshot of their preference for tissue residence or circulation. By comparing IgA- and IgG-dominated clonal groups, we observed subtle differences in the behavior of B cells during ongoing clonal selection and interaction with the tumor. Whereas IgG clonal groups preferentially reside in tumors and draining LNs, IgA-dominated clonal groups show a greater preference for circulation. An obvious question is whether this corresponds to the proportion of isotypes at the cellular level (by surface or intracellular staining). The relative frequencies of antibody-secreting B cells in peripheral blood were IgA1 > IgG1 > IgA2 > IgM > IgG4 > IgG2 > IgG3⁷⁴, which corresponds to our observations of the peripheral blood bulk repertoire (total IgA > total IgG > total IgM). Therefore, the observed clonal group sharing between tissues may simply be due to the higher number of clonotypes, with a certain isotype detected in a certain tissue.

However, for tissue distribution triangle plots, the data were normalized for isotype usage between tissues (i.e., equal numbers of IgG/IgA/IgM most frequent clonotypes were used for PBMC, LN, and blood) to correct for quantitative bias. Therefore, the trend for tissue preference reflects true clonal behavior and not simply quantitative prevalence.

It is well known that only draining LNs show signs of interaction with tumor antigens⁷⁵. In murine models, non-draining usually occurs contralateral to the tumor site, which is anatomically distant. However, whether LNs within the same anatomical site may differ in terms of their interaction with the tumor has yet to be determined. For some of the patients in this study, we were able to obtain more than one lymph node from the same tumor-draining pool, which allowed us to address this question directly on the level of BCR repertoires. A higher overlap between the tumor and LN repertoire signifies a more intensive interaction with the tumor and its antigens. In at least one patient, the LNs studied were significantly different in this regard, with one of the LNs being significantly more similar to the tumor in its BCR repertoire. This observation was also reproduced at the clonal group level, where the lymph node that was in closer interaction with the tumor in terms of clonal overlap also contained more clonal groups that included tumor-derived clonotypes and were confined to this lymph node only. These clonal groups represent antibody maturation and selection processes involving tumors, and occur preferentially in the LN, which is in tight interaction with the tumor, but not in the others. Whether this also reflects the response to tumor antigens and the maturation of tumor-specific antibodies remains to be tested.

It has been previously shown that within a single lymph node, individual germinal centers (GCs) share their immunoglobulin clonal composition to some extent, and the offspring of one B-cell clone may be found in several individual germinal centers⁷⁶. During GC reactions, individual GCs become oligoclonal (containing 4-13 major B cell clones with functional *IgVH*, as detected in the pre-NGS era) due to an affinity selection process⁷⁷⁻⁷⁹. However, in the case of chronic antigen stimulation, as in cancer, repeated cycles of antigen exposure, memory B-cell reactivation,

and germinal center reactions should lead to uniform involvement of the whole LN in interaction with tumors and production of tumor-specific immune responses. Again, we sought to study the spatial heterogeneity of LNs at the level of individual immunoglobulin clonotypes and clonal lineages. We found that the BCR repertoires derived from fragments of a single lymph node were significantly different in their similarity to the pooled tumor BCR repertoire (F2 metric). We conclude that despite chronic antigen exposure, despite the ability of memory B cells to leave and re-enter germinal center reactions and form new germinal centers in subsequent rounds of affinity maturation, the BCR landscape of a lymph node remains significantly heterogeneous in space.

Likewise, for tumor immune infiltrates, it was shown that the TCR repertoire is heterogeneous in space, and this heterogeneity correlates with subclonal mutations within the tumor tissue⁶⁹ and likely represents expanded mutation-specific T cell clones. BCRs are expected to be even more clonal, both in LNs and tumors. However, for BCRs, heterogeneity analysis is complicated by the orders-of-magnitude difference in immunoglobulin expression between plasma cells and other B-cells²¹. A rare plasma cell will have a low probability of sampling at the cellular level but will produce a sufficient amount of immunoglobulin RNA to be reliably detected in the BCR repertoire. Therefore, appropriate use of biological (cellular) replicates sequenced at comparable depths is crucial for accurate detection of differentially expanded clonotypes²³.

Following the logic described above, we were able to analyze repertoire overlap between tumor fragments and detect BCR clonotypes that were differentially expanded in individual fragments. We observed that, in some patients, the most abundant BCR clonotypes were different in different fragments. In this case, the most expanded clonotypes rarely overlapped between fragments, thus probably representing clonotypes that recognize subclonal tumor antigens. This needs to be tested directly and, if proven true, may potentially be a reliable method for the identification of tumor-specific antibodies. In other cases, the tumor may be relatively homogeneous, without significant expansion, and with highly overlapping repertoires of individual fragments. These observations are mostly reproduced at the level of clonal groups, where in heterogeneous tumors, we observed clonal groups that concentrated almost entirely in single tumor fragments, whereas in homogenous tumors, clonal groups mostly include clonotypes derived from all fragments of the tumor.

The physicochemical properties of the CDR-H3 region differ between clonotypes that represent clonal lineages, and thus, are actively hypermutating, and those that do not. The most prominent difference was observed for kf4 (inversely related to hydrophobicity) in PBMC and tumors; clonotypes that undergo hypermutation are, on average, less hydrophobic, which indicates higher binding specificity. However, there was no correlation between kf4 and the extent of hypermutation, even in large full-length Ig-profiling datasets. One possible explanation may be that the selection of less hydrophobic CDR-H3s occurred before the onset of hypermutation.

Interestingly, analysis of silent vs. non-silent hypermutations (dNdS) suggests that in the absence of a defined time-controlled antigenic stimulus, such as vaccination, in steady-state equilibrium, only a few immunoglobulin clonal groups show definitive signs of positive selection. It would be interesting to systematically explore the specificity of these clones in patients with cancer as a potential source of tumor-reactive antibodies.

In general, this study has several limitations, such as the small number of patients, both per cancer type and overall, and our 5'RACE library preparation protocol, which was used for most samples in this study, does not allow for an accurate distinction between the IgA and IgG isotype subtypes. Regarding clonal lineage structure analysis, we found that our study design has a major limitation of a relatively short (150 bp) read length, which does not provide sufficient information for hypermutation analysis.

Nevertheless, our observations contribute to the understanding of the interactions among the tumor immune microenvironment, tumor-draining LNs, and peripheral blood.

Materials and methods

Patients

All clinical samples were acquired from the N.N. Blokhin National Medical Research Center of Oncology or Volga District Medical Centre under the Federal Medical and Biological Agency. This study was conducted in accordance with ICH-GCP. The protocol was approved by the Ethical Committees of the Volga District Medical Centre under the Federal Medical and Biological Agency, and by the N.N. Blokhin National Medical Research Center of Oncology. Written informed consent was obtained from all patients.

Fourteen patients were included in the study, of which four were diagnosed with colon cancer, four with lung adenocarcinoma, and six with melanoma. For each patient, samples from the tumor (tum), draining LN(s) (further referred to as LN1, LN2, LN3 if multiple LNs were analyzed), peripheral blood (PBMC), and normal tissue (norm) were collected, or only some of these tissues (Table S2). Several distant pieces were resected from each tumor and lymph node, unless there were several LNs available (as for colon cancer patients, ccp2, ccp3). All samples were bulk (unsorted), but for two colon cancer patients (ccp2 and ccp3), plasma cells were sorted from a part of each sample. All the samples were processed and stored for library preparation. Blood sampling was performed immediately prior to surgery, and the total volume of obtained blood did not exceed 35 ml for each patient. Fragments of tumor, LN, or normal tissue were placed in 50 ml Falcon tubes containing 7-10 ml of MACS® Tissue Storage Solution (Miltenyi Biotec, cat. 130-100-008) and stored at room temperature for no longer than 2 hours. All the procedures were performed under sterile conditions. PBMC were isolated from the blood samples using the Ficoll-Hypaque™ centrifugation protocol. Briefly, whole blood was diluted 2.5 times with sterile PBS buffer, carefully layered over Ficoll-Paque PLUS (GE Healthcare, cat. 17-1440-03) and centrifuged at 600g for 20 min. Afterwards, buffy coats were harvested from the interface, washed twice with PBS buffer (20 min 350 g), calculated, and lysed in RLT buffer (Qiagen, cat. 79216) at 0.5 to 3 *10⁶ cells/sample density.

Tum, LN, and normal tissues were mechanically cut into 3-7 mm fragments and incubated in RPMI 1640 medium (Gibco™, cat. 42401042) containing 1 mg/ml Liberase TL Research Grade (Roche, cat. 5401020001) and 30 U/ml DNase I (Qiagen, cat. 15200-50), at 37°C for 30–60 min with gentle shaking. Samples were then processed using a gentleMACS™ Dissociator (Miltenyi Biotec, cat. 130-093-235) and passed through a 100 µm Nylon cell strainer to remove non-dissociated fragments. The resulting cell suspension was concentrated by centrifugation (7 min, 350 g) and lysed in RLT buffer (Qiagen, cat. 79216) 0.1-0.5 *10⁶ lymphocytes /sample density. All RLT samples (PBMC, tum, LN, and norm lysates) were stored at -80°C before preparation of the BCR IGH repertoire libraries.

PC isolation

For colon cancer patients, three distant samples (at least 3 cm apart) were obtained from the tumor border with a portion of normal tissue. Cells from three digested tumor samples, three LNs (for patients ccp2 and ccp3) and two replicates of PBMC were stained with a panel of fluorescent antibodies: CD45-PerCP/Cy5.5 (BD 564105, clone HI30), CD38-PE (BD 555460, clone HIT2), CD19-Alexa700 (BD 557921, clone HIB19), CD20-BV510 (BD 563067, clone 2H7), CD25-V450 (BD 560458, clone M-T271) and isolated BD FACSAriaIII cell sorter (BD Bioscience). First, lymphocytes were gated as CD45⁺ cells, and the plasma cell population was isolated as CD20^{neg}/CD19^{low}/CD27^{high}/CD38^{high}. Sorting was performed using a FACSAria III cell sorter (BD Bioscience) directly into RLT lysis buffer (Qiagen, cat 79216). Cells from the ccp6 LNs were lysed and unsorted.

BCR library preparation and sequencing

All BCR IGH libraries were generated using the 5RACE methodology described by Turchaninova et al.²¹ and sequenced with a 150+150 bp read length. For one of the lung cancer patients, lcp3, additional tumor, lymph node, and normal lung tissue libraries were generated using IgMultiplex methodology and sequenced with 250+250 read length. To account for sampling bias, we also obtained technical replicate samples at the cell suspension level.

Preprocessing of sequencing data

To process sequencing reads, we used the MiNN software to extract UMI from raw sequences, correct errors, and assemble consensus sequences for each UMI. For bulk libraries prepared with the 5'RACE protocol and sequenced with 150+150 read length, we filtered out UMI with less than three reads. For bulk libraries prepared with the Multiplex protocol and sequenced with 250+250 read length, we filtered out UMI with less than four reads. For libraries of sorted plasma cells prepared with the 5'RACE protocol and sequenced with 150+150 read length, we filtered out UMI for which there were fewer than two reads.

To ensure the absence of contamination, we checked for the presence of identical UMI in different samples. If such a UMI was identified, and BCR sequences for such UMIs were identical or almost identical (suspecting amplification error), then the number of reads for this UMI in both samples were compared. If the number of reads of this particular UMI in one sample exceeded the number of reads of this UMI in another sample by more than five times, this UMI was eliminated from the sample with a smaller number of reads per UMI. If the ratio was lower than 5, the UMI was eliminated from both samples.

After UMI-based decontamination, we used MIXCR software to assemble reads into quantitated clonotypes, determine germline V, D, and J genes, isotypes, and find the boundaries of target regions, such as CDR-H3. Clonotypes derived from only one UMI were excluded from the analysis of individual clonotype features but were used to analyze clonal lineages and hypermutation phylogeny. Single-UMI sequences were eliminated to avoid errors during cDNA synthesis during library preparation. However, modern reverse transcriptases have high fidelity, with 2.5e-05 error rates for some of them⁸⁰. Therefore, we considered it reasonable to include single UMI sequences in those parts of the analysis, where a larger sample size was important.

Samples with 50 or less clonotypes left after preprocessing were excluded from the analysis.

Clonal lineage inference

We identified sequences belonging to the same clonal lineage using the ChangeO software. The criteria for the initial grouping were the same V and J germline genes identified by MIXCR, and the same CDR-H3 length. These criteria do not account for the D segment, as there is insufficient confidence in the germline annotations due to its short length and high level of mutations. Sequences within each group were defined as belonging to the same clonal lineage if they had a nucleotide CDR-H3 sequence identity above a certain threshold. Such a threshold was individually defined for each patient's dataset as a local minimum of the distance-to-nearest distribution function³¹. In most cases, this threshold is set between 80% and 85%.

Phylogenetic inference

The phylogeny of B-cell hypermutations was inferred for each clonal lineage of size five or more using the maximum likelihood method and the GTR GAMMA nucleotide substitution model. To find the most recent common ancestor (MRCA) or root of the tree, we used an outgroup constructed as a conjugate of germline segments V and J defined by MIXCR. The D segment was

excluded from the outgroup formation, as there was insufficient confidence in the germline annotations due to its short length and high level of mutations. The MUSCLE tool was used for multiple sequence alignment and RAxML software was used to build and root phylogenetic trees.

Sample pooling and normalization

For some parts of the analysis, we considered samples irrespective of technical replicates or specific tumor/LN segments. In such cases, the corresponding repertoire datasets were pooled in one, where for each clonotype, its new frequency was calculated as the mean of the clonotype's frequencies from separate files. Such an approach helps avoid sampling bias and achieve equal contributions of all clonotype sets being pooled.

Statistical analysis

In our analysis, we often used repertoires of different parts of the same tissue as separate observations within the same comparison. Examples of these could be isotypes, different pieces of tumors or LNs, or PBMC samples taken at different time points. Understanding the pitfalls of this approach in general, we argue that it can be justified in some cases considering the heterogeneity of tissues, especially tumors, and the distinctive characteristics of different isotypes.

Most of the analysis was performed using VDJtools and custom Python and R scripts.

Expanded clonotypes detection

We used the EdgeR package in R to identify the clonotypes that were differentially expressed between the two sample sets. The problem with this approach is determining the correct number of counts required to pass into the DGEList function of EdgeR. Using a number of unique UMIs detected for each clonotype in the sample might not be a good idea, considering the possibility of sampling bias (e.g., resecting tumors into pieces of slightly different sizes). To account for sampling bias, we defined clonotype count for the DGEList function as clonotype frequency in a normalized sample multiplied by the total number of unique UMIs in all groups of samples.

The output of the DGEList function is then normalized and passed to the exactTest function of the EdgeR. Clonotypes with $FDR < 0.05$ and $\log FC > 0$ were considered expanded in a corresponding group of samples.

Immunohistochemistry

Sections of formalin-fixed paraffin-embedded tissue were sliced on an RM2235 microtome (Leica Biosystems). Slices were deparaffinized with Xylol and Ethanol. H&E staining was performed using Mayer's hematoxylin and eosin (Biovitrum, Russia). For IHC staining, the slices were demasked in AR9 buffer (PerkinElmer) for 10 min at 98 °C in a water bath, washed twice in PBS, and blocked for 30 min with Protein Block (Leica Novocastra, UK). Primary antibodies were added without washing the blocking solution and incubated overnight at 4 °C. After incubation with primary antibodies, slices were washed twice in PBS and stained with the NovoLink polymer detection system (Leica Biosystems, UK) according to the manufacturer's instructions or with HRP-streptavidin (Jackson ImmunoResearch 016-620-084, 1:500 for 30 min) and contrasted with 10 µM CF tyramide dye (Biotium, USA). The following primary antibodies were used at the consequence and with indicated dilutions and CF dyes: CD20 (Leica Biosystems PA0200), 1:1 with CF488; CXCL13 (GeneTex GTX108471, Taiwan), 1:100 with CF660; CD3 (HuaBio HA720082, China), 1:200 with CF430; PNAd-biotin (BioLegend 120804, USA), 1:200 with CF555. Before staining with non-biotinylated antibodies, the antibodies were stripped in AR9 buffer for 10 min at 98 °C. After staining with antibodies, the slides were counterstained with DAPI (0.2 nM) for 5 min, embedded in Mowiol 4-88 (Sigma-Aldrich), and coverslipped.

Brightfield and fluorescence images were acquired using an EVOS M7000 microscope (Thermo Fisher Scientific) with a 20x dry objective. The following light cubes were used: DAPI (AMEP4650) for DAPI fluorescence, CFP (AMEP4653) for CF430 fluorescence, YFP (AMEP4654) for CF488 fluorescence, RFP (AMEP4652) for CF555 fluorescence, and Cy5 (AMEP4656) for CF660. Whole slice images were stitched using microscope imaging software. Fluorescence images were contrasted, colorized into pseudocolors, and overlaid using Fiji software (<https://imagej.net/>.

Abbreviations

- (PBMC): Peripheral blood mononuclear cells
- (LN): lymph node
- ((tumor)-draining LNs): (tumor)-draining lymph nodes
- (BCR): B cell receptor
- (CDR-H3): complementarity-determining region 3 of a BCR heavy chain
- (TME): tumor microenvironment
- (SHM): somatic hypermutation
- (TLSs): tertiary lymphoid structures
- (TI-Bs): tumor-infiltrating B-cells
- (ADCC): antibody-dependent cytotoxicity
- (MDSC): myeloid-derived suppressor cells
- (LUAD): lung adenocarcinoma
- (**R**apid **A**mplification of **c**DNA **E**nds): 5'RACE
- (IGH): immunoglobulin heavy chain
- (SLE): systemic lupus erythematosus
- (CAR-T): chimeric antigen receptor T cell
- (MRCA): most recent common ancestor.

Terms “tumor-draining LN”, “draining LN” and “LN” are used interchangeably throughout the text and figures.

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Reviewer #3 (Public Review):

In multiple cancers, the key roles of B cells are emerging in the tumor microenvironment (TME). The authors of this study appropriately introduce that B cells are relatively under-characterised in the TME and argue correctly that it is not known how the B cell receptor (BCR) repertoires across tumor, lymph node and peripheral blood relate. The authors therefore supply a potentially useful study evaluating the tumor, lymph node and peripheral blood BCR repertoires and site-to-site as well as intra-site relationships. The authors employ sophisticated analysis techniques, although the description of the methods is incomplete.

Major strengths:

(1) The authors provide a unique analysis of BCR repertoires across tumor, dLN, and peripheral blood. The work provides useful insights into inter- and intra-site BCR repertoire heterogeneity. While patient-to-patient variation is expected, the findings with regard to intra-tumor and intra-dLN heterogeneity with the use of fragments from the same tissue are of importance, contribute to the understanding of the TME, and will inform future study design.

(2) A particular strength of the study is the detailed CDR3 physicochemical properties analysis which leads the authors to observations that suggest a less-specific BCR repertoire of TIL-B compared to circulating B cells.

Concerns and comments on current version:

The revision has improved the manuscript but, in my opinion, remains inadequate. While most of my requested changes have been made, I do not see an expansion of Fig1A legend to incorporate more details about the analysis. Lacking details of methodology was a concern from all reviewers. Similarly, the 'fragmented' narrative was a concern of all reviewers. These matters have not been dealt with adequately enough - there are parts of the manuscript which remain fragmented and confusing. The narrative and analysis does not explain how the plasma cell bias has been dealt with adequately and in fact is simply just confusing. There is a paragraph at the beginning of the discussion re the plasma cell bias, which should be re-written to be clearer and moved to have a prominent place early in the results. Why are these results not properly presented? They are key for interpretation of the manuscript. Furthermore, the sorted plasma cell sequencing analysis also has only been performed on two patients. Another issue is that some disease cohorts are entirely composed of patients with metastasis, some without but metastasis is not mentioned. Metastasis has been shown to impact the immune landscape.

A reviewer brought up a concern about the overlap analysis and I also asked for an explanation on why this F2 metric chosen. Part of the rebuttal argues that another metric was explored showing similar results, thus conclusion reached is reasonable. Remarkably, these data are not only omitted from the manuscript, but is not even provided for the reviewers.

This manuscript certainly includes some interesting and useful work. Unfortunately, a comprehensive re-write was required to make the work much clearer and easier to understand and this has not been realised.

<https://doi.org/10.7554/eLife.89506.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors attempt to fully characterize the immunoglobulin (Ig) heavy (H) chain repertoire of tumor-infiltrating B cells from three different cancer types by identifying the IgH repertoire overlap between these, their corresponding draining lymph nodes (DLNs), and peripheral B cells. The authors claim that B cells from tumors and DLNs have a closer IgH profile than those in peripheral blood and that DLNs are differentially involved with tumor B cells. The claim that tumor-resident B cells are more immature and less specific is made based on the characteristics of the CDR-H3 they express.

Strengths:

The authors show great expertise in developing in-house bioinformatics pipelines, as well as using tools developed by others, to explore the IgH repertoire expressed by B cells as a means of better characterizing tumor-associated B cells for the future generation of tumor-reactive antibodies as a therapy.

Weaknesses:

This paper needs major editing, both of the text and the figures, because as it stands it is convoluted and extremely difficult to follow. The conclusions reached are often not obvious from the figures themselves. Sufficient a priori details describing the framework for their analyses are not provided, making the outcome of their results questionable and leaving the reader wondering whether the findings are on solid ground.

The authors are encouraged to explain in more detail the premises used in their algorithms, as well as the criteria they follow to define clonotypes, clonal groups, and clonal lineages, which are currently poorly defined and are crucial elements that may influence their results and conclusions.

In response to this comment, we significantly expanded the paragraph dedicated to the tumor and non-tumor repertoire overlap and isotype composition. The following sections were added:

First, we characterized the relative similarity of IGH repertoires derived from tumors, DLN, and PBMC on the individual CDR-H3 clonotype level. We define clonotype as an instance with an identical CDR-H3 nucleotide sequence and identical V- and J- segment attribution (isotype attribution may be different). Unlike other authors, here we do not pool together similar CDR-H3 sequences to account for hypermutation. (Hypermutation analysis is done separately and defined as clonal group analysis.)

As overlap metrics are dependent on overall repertoire richness, we normalized the comparison using the same number of top most frequent clonotypes of each isotype from each sample ($N = 109$). Repertoire data for each sample were split according to the immunoglobulin isotype, and the F2 metric was calculated for each isotype separately and plotted as an individual point.

We also analyzed D metric, which represents the relative overlap diversity uninfluenced by clonotype frequency ($D_{ij} = d_{ij} / (d_i * d_j)$, where d_{ij} is the number of clonotypes present in both samples, while d_i and d_j are the diversities of samples i and j respectively). The results for D metric are not shown, as they indicate a similar trend to that of F2 metric. This observation allows us to conclude that tumor IGH repertoires are more similar to the repertoires of lymph nodes than to those of peripheral blood, both if clonotype frequency is taken into account, and when it is not.

Having excluded the IGHD gene segment from some of their analyses (at least those related to clonal lineage inference and phylogenetic trees), it is not well explained which region of CDR-H3 is responsible for the charge, interaction strength, and Kidera factors, since in some cases the authors mention that the central part of CDR-H3 consists of five amino acids and in others of seven amino acids.

We considered different ways of calculating amino acid properties of CDR3 and used different parameters for sample-average and individual-sequence CDR3s. Now plots for Fig S6 C are updated for consistency and the parameters depicted there are now calculated using 5 central amino acids, as in other sections.

How can the authors justify that the threshold for CDR-H3 identity varies according to individual patient data?

Ideal similarity threshold may depend on several factors, such as sampling, sequencing depth etc. For example, imagine a sample picking up 100% of the clonal lineage sequences which differ only 1 amino acid from each other, and a worse quality sample/sequencing picking up only every other sequence. Obviously, the minimal threshold required to accumulate these into a cluster/clonal group would be different for these two cases (1aa for the former, and ~2 aa for the latter for single-linkage clustering). Or, in other words, the more the sequencing depth, the more dense the clusters will be. The method of individual threshold tailoring relies on the following: <https://changeo.readthedocs.io/en/latest/examples/cloning.html>

Although individual kidera factors that are significant in the context of our analysis are described in the text one by one on their first appearance, we now also added a sentence to describe Kidera factor analysis in general (page 8):

Kidera factors are a set of scores which quantify physicochemical properties of protein sequences (Nakai et al. 1988). 188 physical properties of the 20 amino acids are encoded using dimension reduction techniques.

Throughout the analyses, the reasons for choosing one type of cancer over another sometimes seem subjective and are not well justified in the text.

Whenever possible, we pooled all patients with all cancer types together, because the number of available samples did not allow us to draw any significant conclusions comparing between individual cancer types. When analyzing and showing individual patient data, we also did not attempt to depict any cancer-type-specific findings, but it is inevitable that we name a specific cancer type when labelling a sample coming from a specific tumor.

Overall, the narrative is fragmented. There is a lack of well-defined conclusions at the end of the results subheadings.

In addition to the described above, a conclusion was added to the paragraph describing hypermutation analysis:

IGHG clonotypes from lung cancer samples show higher number of hypermutations, possibly reflecting high mutational load found in lung cancer tissue. For melanoma, another cancer known for high mutational load, no statistically significant difference was found. This may be due to higher variance between melanoma samples, which hinders the analysis, or due to the small sample size.

The exact same paragraph is repeated twice in the results section.

Corrected.

The authors have also failed to synchronise the actual number of main figures with the text, and some panels are included in the main figures that are neither described nor mentioned in the text (Venn diagram Fig. 2A and phylogenetic tree Fig. 5D). Overall, the manuscript appears to have been rushed and not thoroughly read before submission.

Corrected.

Reviewers are forced to wade through, unravel, and validate poorly explained algorithms in order to understand the authors' often bold conclusions.

We hope that the aforementioned additions to the text and also addition to the Figure 1 make the narrative more easily understandable.

Reviewer #2 (Public Review):

Summary:

The authors sampled the B cell receptor repertoires of Cancers, their draining lymph nodes, and blood. They characterized the clonal makeup of all B cells sampled and then analyzed these clones to identify clonal overlap between tissues and clonal activation as expressed by their mutation level and CDR3 amino acid characteristics and length. They conclude that B cell clones from the Tumor interact more with their draining lymph node than with the blood and that there is less mutation/expansion/activation of B cell clones in Tumors. These conclusions are interesting but hard to verify due to the under-sampling and short sequencing reads as well as confusion as to when analysis is across all individuals or of select individuals.

Strengths:

The main strength of their analysis is that they take into account multiple characteristics of clonal expansion and activation and their different modes of visualization, especially of clonal expansion and overlap. The triangle plots once one gets used to them are very nice.

Weaknesses:

The data used appears inadequate for the conclusions reached. The authors' sample size of B cells is small and they do not address how it could be sufficient. At such low sampling rates, compounded by the plasmablast bias they mention, it is unclear if the

overlap trends they observe show real trends. Analyzing only top clones by size does not solve this issue. As it could be that the top 100 clones of one tissue are much bigger than those of another and that all overlap trends are simply because the clones are bigger in one tissue or the other. i.e there is equal overlap of clones with blood but blood is not sufficiently sampled given its greater diversity and smaller clones.

Regarding the number of clonotypes to be taken into account, we were limited by the B cell infiltration of tumor samples and our ability to capture their repertoire. However, we use technical replicates on the level of cell suspension to ensure that at least top clonotypes are consistently sampled. So, this is how the data should be interpreted - as describing the most abundant clones in the repertoire (which also may be considered the most functionally relevant in case of tumor infiltrating lymphocytes).

To analyze the repertoire overlap, we generally use the F2 metric that takes clone size into account - because we think that clone size is an important functional factor. However, we have now added the description of using D metric (does not include clone frequency as a parameter) - which shows exactly the same trend as F2 metric. So, both F2 and D overlap metrics support our conclusion of higher overlap between tumor and LN.

The following text was added:

We also analyzed D metric, which represents the relative overlap diversity uninfluenced by clonotype frequency ($D_{ij} = d_{ij} / (d_i * d_j)$, where d_{ij} is the number of clonotypes present in both samples, while d_i and d_j are the diversities of samples i and j respectively). The results for D metric are not shown, as they indicate a similar trend to that of F2 metric. This observation allows us to conclude that tumor IGH repertoires are more similar to the repertoires of lymph nodes than to those of peripheral blood, both if clonotype frequency is taken into account, and when it is not.

All in all, of course, the deeper the better, but given the data we were able to generate from the samples, this was the best approach to normalization that could be used.

Similarly, the read length (150bp X2) is too short, missing FWR1 and CDR1 and often parts of FWR2 if CDR3 is long. As the authors themselves note (and as was shown in (Zhang 2015 - PMC4811607) this makes mutation analysis difficult.

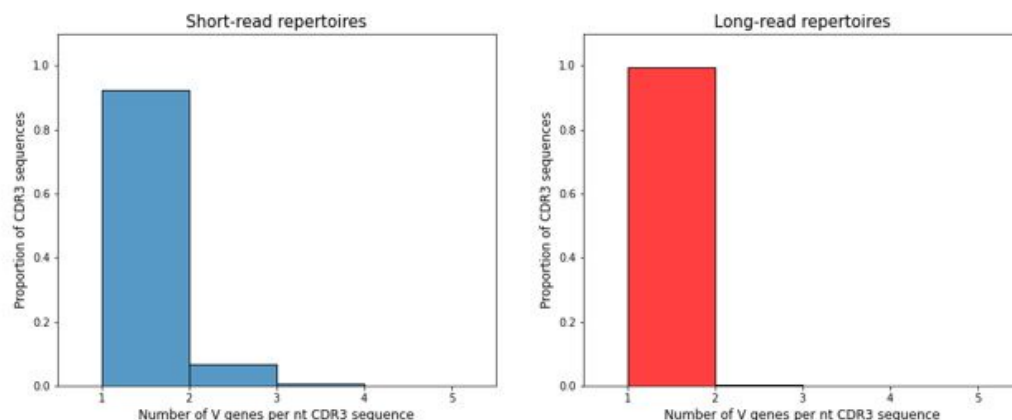
Indeed, we are aware of this problem, and therefore only a small part of the manuscript is dedicated to the hypermutation analysis. However, as the CDR-H3 region is the most mutated part, we still can capture significant diversity of mutations. To address the question of applicability of our data for the hypermutation phylogeny analysis, we compare the distribution of physico-chemical properties along the trees of hypermutation using the 150+150 and 300+300 data from the same donor and the same set of samples. The main conclusion is that neither for long, nor for short datasets could any correlation of physicochemical properties of the CDR-H3 region with the rank of the clonotype on the tree be found.

It also makes the identification of V genes and thus clonal identification ambiguous. This issue becomes especially egregious when clones are mutated.

Again, this would be important for clonotype phylogeny analysis. However, for the simple questions that we address with our clonal group analysis, such as clonal group overlap between tissues etc, we consider this data acceptable, because if any mislabelling of V segment occurs, it is a) rare and b) is equally frequent in all types of samples. Therefore, any conclusions made are still valid despite this technical drawback.

To directly address the question of mislabelling of V-genes in our data, we looked at the average number of different V-genes attributed to the same nucleotide sequence of CDR-H3 region in the short (150+150) and long (300+300) datasets from the same donor. Indeed, some ambiguity of V-gene labelling is observed (see below), but we think that it is unlikely to influence any of our cautious conclusions.

Author response image 1.



Finally, it is not completely clear when the analysis is of single individuals or across all individuals. If it is the former the authors did not explain how they chose the individuals analyzed and if the latter then it is not clear from the figures which measurements belong to which individual (i.e they are mixing measurements from different people).

We addressed this issue by adding a comment to each figure caption, describing whether a particular figure or panel describes individual or pooled data, and also whether the analysis is done on individual clonotype or clonal group level.

Also, in case pooled data were used, we added the number of patients that was pooled for a particular type of analysis. This number differs from one type of analysis to the other, because not all the patients had a complete set of tissues, and also not all samples passed a quality check for a particular analysis.

Here are the numbers listed:

Fig 2A: N=6 (we were only considering those who had all three tissues)

Fig 2C, N=14 (all)

2D: N=14 (all)

2E N=7 (have both tum and PBMC).

2F N=9 (have both tum and PBMC).

2G N=9 (have both tum and PBMC)

2H N=7 (have both tum and LN)

3A N=14 (all)

3B N=11 (only those with tumor)

3E - N=14

7F N=11 (all that have tumor)

Reviewer #3 (Public Review):

In multiple cancers, the key roles of B cells are emerging in the tumor microenvironment (TME). The authors of this study appropriately introduce that B cells are relatively under-characterised in the TME and argue correctly that it is not known how the B cell receptor (BCR) repertoires across tumors, lymph nodes, and peripheral blood relate. The authors therefore supply a potentially useful study evaluating the tumor, lymph node, and peripheral blood BCR repertoires and site-to-site as well as intra-site relationships. The authors employ sophisticated analysis techniques, although the description of the methods is incomplete. Among other interesting observations, the authors argue that the tumor BCR repertoire is more closely related to that of draining lymph node (dLN) than the peripheral blood in terms of clonal and isotype composition. Furthermore, the author's findings suggest that tumor-infiltrating B cells (TIL-B) exhibit a less mature and less specific BCR repertoire compared with circulating B cells. Overall, this is a potentially useful work that would be of interest to both medical and computational biologists working across cancer. However, there are aspects of the work that would have benefitted from further analysis and areas of the manuscript that could be written more clearly and proofread in further detail.

Major Strengths:

(1) The authors provide a unique analysis of BCR repertoires across tumor, dLN, and peripheral blood. The work provides useful insights into inter- and intra-site BCR repertoire heterogeneity. While patient-to-patient variation is expected, the findings with regard to intra-tumor and intra-dLN heterogeneity with the use of fragments from the same tissue are of importance, contribute to the understanding of the TME, and will inform future study design.

(2) A particular strength of the study is the detailed CDR3 physicochemical properties analysis which leads the authors to observations that suggest a less-specific BCR repertoire of TIL-B compared to circulating B cells.

Major Weaknesses:

The study would have benefitted from a deeper biological interpretation of the data. While given the low number of patients one can plausibly understand a reluctance to speculate about clinical details, there is limited discussion about what may contribute to observed heterogeneity.

We indeed do not want to overinterpret our data, especially where it comes to the difference between types of cancer. On the other hand, extracting similar patterns between different cancer types allows to pinpoint mechanisms that are more general and do not depend on cancer type. As for the potential source of intratumoral heterogeneity that we observe, we think that it may be coming from the selective sampling of tertiary lymphoid structures. We include IHC data for TLS detection in the supplementary Fig.5. Also, tumor mutation clonality may correlate with differential antibody response (i.e. different IGH clonotypes developing to recognize different antigens) – as has been previously described for TCRs by the lab of B.Chain in <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6890490/>.

For example, for the analysis of three lymph nodes taken per patient which were examined for inter-LN heterogeneity, there is a lack of information regarding these lymph nodes.

Unfortunately no clinical information about the lymph nodes was available.

'LN3' is deemed as exhibiting the most repertoire overlap with the tumor but there is no discussion as to why this may be the case.

The following phrases describes this in the “LN-to-LN heterogeneity in colorectal cancer” paragraph:

Similarly, an unequal interaction of tumors with DLNs was observed at the level of hypermutating clonal groups.

Functionally, this may again indicate that within a group of DLNs, nodes are unequal in terms of access to tumor antigens, and this inequality shapes the BCR repertoires within these lymph nodes.

(2) At times the manuscript is difficult to follow. In particular, the 'Intra-LN heterogeneity' section follows the 'LN-LN heterogeneity in colorectal cancer' section and compares the overlap of LN fragments (LN11, LN21, LN31) with the tumor in two separate patients (Fig 6A). In the previous section (LN-LN), LN11, LN21, LN31 are names given to separate lymph nodes from the same patient. The fragments are referred to as 'LN2' and the nodes in the previous section are referred to similarly. This conflation of naming for nodes and fragments is confusing.

We corrected this.

(3) There is a duplicated paragraph in 'Short vs long trees' and the following section 'Productive involvement in hypermutation lineages depends on CDR3 characteristics.

Corrected.

Reviewer #1 (Recommendations For The Authors):

- Figures:

Figure 1A lacks resolution

Corrected

Figure 2A, Venn diagram: What do the colors indicate?

Corrected

Figure 5D, why include this tree when there is no mention of it in the text?

Described

Figures 8, 9, and 10 are not to be found. One should not have to figure out that they became supplementary in the end.

Corrected

Regarding the physicochemical properties of CDR-H3, what do the authors mean by "the central part"? Do the authors refer to the CDR-H3 loop, and if so, how is that defined when the IGHD gene segment is excluded from the analyses? Is it 5 amino acids (Productive involvement in hypermutating lineages depends on CDR3 characteristics, Page 21/39 in merged document) and (CDR3 properties, Page 8/39 in merged document), or 7 amino acids (Short vs long trees phylogeny analysis, Page 19/39 in merged document)? Please clarify.

We considered different ways of calculating amino acid properties of CDR3 and used different parameters for sample-average and individual-sequence CDR3s. Now plots for Fig S6 C are updated for consistency. IGHD segment was not excluded from the analysis. The reviewer might be confused by our description of phylogenetic inference, when an artificial outgroup with D segment deleted is added to the clonal group to facilitate the inference process. All other sequences were analyzed in their original form with the D segment. This way, we could avoid biases in phylogeny introduced by misassignment of D gene germline to the outgroup.

What was the threshold for CDR-H3 identity in their analyses? How can the authors justify that this value changes according to individual patient datasets? (Materials & methods, Clonal lineage inference Page 29/39 in merged document).

As described earlier, ideal similarity threshold may depend on several factors, such as sampling, sequencing depth etc. For example, imagine a sample picking up 100% of the clonal lineage sequences which differ only 1 amino acid from each other, and a worse quality sample/sequencing picking up only every other sequence. Obviously, the minimal threshold required to accumulate these into a clonotype would be different for these two cases (1aa for the former, and ~2 aa for the latter for single-linkage clustering). The method of individual threshold tailoring relies on this: <https://changeo.readthedocs.io/en/latest/examples/cloning.html>

What is the difference between tumor-induced and tumor-infiltrating B cells? How can the authors discriminate between the two? Page 6/39 in the merged document.

corrected to tumor-infiltrating

"Added nucleotides" meaning N additions? Page 3/39 in the merged document.

yes

How many cancer patients were enrolled? 17 or 14(Materials & methods page 27/39 in the merged document)? Please clarify.

In the current project 14 patients were enrolled. The appropriate changes have been introduced in the final text. Supplementary table 2 has been added with the patient data.

Abbreviations are used without full descriptions.

According to reviewer's recommendation, a list of abbreviations was added in the manuscript, and also full descriptions were added in the text upon first mentioning of the term.

Use either CDR3 or CDR-H3

We corrected the text to use CDR-H3 abbreviation throughout the text.

Reviewer #2 (Recommendations For The Authors):

I would like to start by apologizing for the time it took me to review.

As I mentioned above there are issues with the clonal sampling of the sequencing length and the statistics in this paper. From reading the paper I am not sure if they are fixable but there are some things that could be tried.

(1) The authors mention the diversity of their individual analysis - 17 individuals across 3 cancer types, but do not then systematically show us how the different things they measure track across the different individuals and cancer types. it is possible that some trends would be more convincing if we saw them happening again and again across all individuals. But, as I said above, the authors do not identify individuals clearly across all their types of analysis nor do they explain why sometimes they show analysis of specific individuals.

For overlap analysis (Fig. 2 except panel B), CDR3 properties analysis (Fig. 3, Fig. S7), clonal group analysis (Fig. 4) we used pooled data on all cancers, unless it is indicated otherwise on the panel. For overlap analysis, we used Cytoscape graph (Fig. 2B) for one patient, mp3, to illustrate the findings that were made on pooled data. For other types of analysis, such as overlap between individual lymph nodes, or tumor fragments (Fig. 5, 6, 7 except panel F) pooled analysis is not possible due to the individual nature of the processes in question.

(2) The authors do not address how lacking their sampling is nor the distribution of clone sizes in different tissues/ individuals/ subsets. Without such a discussion it is not clear how tenuous or convincing their conclusions are.

(3) The short sequencing lengths limit the ability to exactly identify V and thus the germline root of clones, whose positions are mutated and clonal association of sequences. The authors appear to be aware of this as they often use the most common ancestor as the start of their analysis... however, again there are inconsistencies that are not clearly described in the text. in creating trees with change they defined roots as the putative germline and at least in most cases also in clone association although in some analyses potentially similar clones were collapsed into clonotypes. Again it is not clear when one method was used or the other and how the choice was made what to choose.

Here we can only state that we consistently used the approach described in the Methods section, which was the following:

First, the repertoires were clustered into clonal lineages using the criteria described in “Methods: Clonal lineage inference” Assuming that each clonotype sequence in the clonal lineage originated from the same ancestor, we try to recover the phylogeny. Please note that we refer to the individual BCR sequences as “clonotypes”, and to a group of clonotypes that presumably share a common ancestor - as “clonal lineage” or “clonal group”.

The phylogeny of B-cell hypermutations was inferred for each clonal lineage of size five or more using the maximum likelihood method and the GTR GAMMA nucleotide substitution model. To find the most recent common ancestor (MRCA) or “root” of the tree, we used an artificial outgroup constructed as a conjugate of germline segments V and J defined by MIXCR and added it to the clonal lineage. The D segment was excluded from the outgroup formation, as there was insufficient confidence in the germline annotations due to its short length and high level of mutations. The rest of the clonotypes were still analyzed in their original form with D segment in place. Deleting D segment from the outgroup simply eliminates the risk of

biasing the phylogeny by missigning D segment germline sequence to the outgroup. The MUSCLE tool was used for multiple sequence alignment and RAxML software was used to build and root phylogenetic trees.

(4) Beyond the statistical issues mentioned above: the unclear selection of individual examples for comparison and significance testing, the mixing of individuals and cancer types without clear identification, etc. there is in general a lack of coherence in the statistical analysis performed. specifically:

(a) the authors should choose one cutoff for significance (0.01 for instance) and then just mention when things are significant and when not. There is no need and it is confusing to add the p-value for every comparison. P-values are not good measures of effect size.

We corrected the figures and left p-values only where they are below significance threshold.

(b) the Bonferroni correction used is not well characterized. For an alpha of 0.01 in Figures 3 C and D how many tests were performed?

The number of tests performed that was used for Bonferroni-Holm correction equals the number of comparisons on the heatmap which makes it 39 for each heatmap on Fig 3C and 13 for Fig 3D.

Finally some minor issues -

(1) Not all acronyms are described, for instance, TME and TIL. The first time any acronym is used it should be spelled out. -> Katya B- спусок сокращений

(2) The figure captions are not all there...

(a) there is no caption for Figure 3E.

corrected

(b) there are Figure 7 F and G panels but no Figure 7E panel and Figure F is described after Figure G.

corrected

(3) A few problems with wording -

(a) bottom paragraph of page 3 - instead of:

"different lymph nodes from one draining lymph node pool may be more or less involved"

Corrected to "different lymph nodes from one draining lymph node pool may be differentially involved"

(b) figure caption for figure 3a: instead of:

"CDR3 are on average significantly higher in tumor"

Corrected to "CDR3 are on average significantly longer in tumor"

Reviewer #3 (Recommendations For The Authors):

- FIG1A - Suggest expanding the legend to include more information on the computational analyses.

added

- PAGE SIX: Suggest adding a table or some text on patient characteristics. Numbers of unique clonotypes per sample etc. Are there differences in age/sex that need to be considered? Some clonotype information is available in S1 but some summary and statistics would be appreciated.

Added patient information as Supplementary table 2.

- PAGE SIX: F2 Metric, suggestion to explain why this was used vs. other metrics.

We expanded the following paragraph to include information about F2 metric and D metric, and the reason why we are using F2.

Repertoire data for each sample were split according to the immunoglobulin isotype, and the F2 metric was calculated for each isotype separately and plotted as an individual point. We used the repertoire overlap metric F2 (Clonotype-wise sum of geometric mean frequencies of overlapping clonotypes), which accounts for both the number and frequency of overlapping clonotypes (Fig. 2A). As expected, significantly lower overlaps were observed between the IGH repertoires of peripheral blood and tumors compared to LN/tumor overlaps. The LN/PBMC overlap also tended to be lower, but the difference was not statistically significant. We also analyzed D metric, which represents the relative overlap diversity uninfluenced by clonotype frequency ($D_{ij} = d_{ij} / (d_i * d_j)$, where d_{ij} is the number of clonotypes present in both samples, while d_i and d_j are the diversities of samples i and j respectively). The results for D metric are not shown, as they indicate a similar trend to that of F2 metric. This observation allows us to conclude that tumor IGH repertoires are more similar to the repertoires of tumor-draining LNs than to those of peripheral blood, both if clonotype frequency is taken into account, and when it is not.

- PAGE SIX: Make clear in the text that mp3 is a patient.

Added “melanoma patient mp3”

- PAGE EIGHT: Suggest explaining kidera factors at first use - not all readers will know what they are.

We expanded the following paragraph to add more information about Kidera factors:

To explore CDR-H3 physicochemical properties, we calculated the mean charge, hydropathy, predicted interaction strength, and Kidera factors 1 - 9 (kf1-kf9) for five central amino acids of the CDR-H3 region for the 100 most frequent clonotypes of each sample using VDJtools. Kidera factors are a set of scores which quantify physicochemical properties of protein sequences [61](#). 188 physical properties of the 20 amino acids are encoded using dimension reduction techniques, to yield 9 factors which are used to quantitatively characterize physicochemical properties of amino acid sequences.

- Fig 5D is not referred to.

Corrected

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